

Laboratory Protocol Handbook

Oladokun Laboratory

Poultry Microbiome and Health Research Group
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oladokunlab.com

A consolidated handbook of training requirements, general laboratory practice, cell culture, extracellular vesicle isolation, and in vitro assay protocols.

IMPORTANT

This handbook supplements — but does not replace — institution-approved biosafety, chemical-safety, organism-specific, equipment and waste-disposal procedures.

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CHAPTER

01

Student Lab Training & Onboarding

Revised April 2025 by IWO

Student Name: _____

OBJECTIVE

To ensure that all students in the Oladokun Lab receive the required training, and to verify that training was received. The student initials each item as the training is completed.

1.1 Required Institutional Training

TAMU-required training must be completed **prior to entry into the lab**. Non-state employees access training through Gateway — set this up at the in-person BSL-2 training using your email address.

#	COURSE	CODE / PLATFORM
☐	BSL-2 Training — Biosafety	211486 · TrainTraq
☐	Hazard Communication	11020 · TrainTraq
☐	Biological Safety Cabinet	2111531 · TrainTraq
☐	Occupational Health and Safety	www.bohp.tamu.edu → "Questionnaire" (NetID login)
☐	Pandemics: Slowing the Spread	Bioraft
☐	Working Safely with Cryogenics	211228 · TrainTraq
☐	Laboratory Safety Training (Online) — EHS	2114106 · TrainTraq
☐	Bloodborne Pathogens for Research Personnel	2114036 · BOHP TrainTraq
☐	Effective Use of Class 2 Biosafety Cabinet	2111531 · TrainTraq
☐	Animal Allergen	2113938 · BOHP TrainTraq
☐	Working with the IACUC	CITI
☐	Working with Poultry in Agricultural Research Settings	CITI
☐	Waste Anesthetic Gas (WAG) Awareness	2114681 · TrainTraq

1.2 Equipment Proficiency

Prior to using laboratory equipment, manuals and laboratory protocols will be read and **proficiency demonstrated to the PI.**

#	EQUIPMENT
☐	Autoclave
☐	Centrifuge
☐	Hot / Stir Plates
☐	Water Bath
☐	Reverse Osmosis Water System
☐	Biological Safety Cabinet
☐	Incubators
☐	Pipettors
☐	Full Face Mask Respirators

1.3 Lab Safety Commitments

Initial each statement.

- ☐ **Closed-toed shoes and a lab coat** will always be worn in the lab.
- ☐ **Gloves and lab coats will not be worn outside the laboratory**, unless safety precludes any risk of contamination — such as moving autoclaved materials or biohazardous waste.
- ☐ **Safety goggles** will be worn when a splash hazard exists.
- ☐ **Fire extinguisher, spill kit, drench hose/eye wash, electrical shut-off and exits** will be identified and their operation understood.
- ☐ **Broken glass** will be disposed of in a broken glass container, never the regular trash.
- ☐ **Sharps** (needles, scalpels) will be disposed of in a sharps container.

- ☐ **All biologicals** will be treated by autoclave (preferred) or chemical disinfection before leaving the building. Solid autoclaved waste will be crossed with an "X" of autoclave tape over the biohazard label, labeled as treated, bagged in an opaque trash bag, and placed in the outside dumpster.
- ☐ **Hands will be washed frequently.** Be aware of cross-contamination of surfaces; do not inadvertently transfer pathogens or ingest biologicals by touching your face or personal items. Personal items go in a marked "clean area" and are not used while working.
- ☐ **Laboratory injuries or possible laboratory-acquired infections** will be reported to the PI immediately.
- ☐ **All surfaces** will be cleaned and disinfected after use with **70%** ethanol, **10%** bleach, quaternary ammonia disinfectant, or 409.
- ☐ **"Dirty" items** will be placed in their designated areas to prevent cross-contamination.
- ☐ **Flammable items** will be placed into the Flammable Cabinet when not in use.
- ☐ **Acids** will be separated in the Acid Cabinet, placed in tubs with absorbent material to contain spills.
- ☐ **All chemicals and reagents** will be marked with received date and opened date, with the initials of the person receiving or opening the product.
- ☐ **TAMU-EHS procedures** will be followed for handling and disposing of chemicals and reagents.
- ☐ **All samples** will be identified by a simple code understandable by all. A paper copy of the code shall be placed in a waterproof bag within the sample box. Samples must include collection date and owner's initials. **All unlabeled samples will be disposed of by the PI as biohazardous waste.**
- ☐ **All filenames** follow the format: *author last name, year of creation, short 3–5 word title, date*. Additions such as "rv by SO" may indicate who reviewed the document.
- ☐ **Copies of data files** will be forwarded to the PI as they are produced — to keep everyone in the loop and to serve as backup.
- ☐ **All electronic files will be backed up:** your computer, your personal backup drive, the PI's computer, and possibly a lab mate's computer. Paper copies of raw data are encouraged.

1.4 Housekeeping

- ☐ Bench tops cleaned and disinfected; floors swept and mopped with Lysol floor cleaner **weekly**.
- ☐ **Laboratory notebooks** kept by each student, inspected periodically by the PI for completeness. Notebooks are the property of the PI and TAMU and **will not leave the laboratory**.
- ☐ **No solids, biologicals, solvents or chemicals** down the drains — including floor dust, liquid agar, or eggshells.
- ☐ Each student enters the lab **at least once per workday** to verify freezers and refrigerators are working, and corrects anything needing improvement (trash on floor, dirty dishes, etc.).
- ☐ **Eye wash station tested weekly**, verified with date and initials.
- ☐ **Shopping list** maintained on the laboratory blackboard. Purchases typically made every 1–2 months; quantities and catalog numbers required.
- ☐ Students read, understand and follow the Oladokun Laboratory **Lab Safety Manual** and protocols (separate files).

1.5 Acknowledgement

I understand that failure to comply with these rules can cause injury to myself or my colleagues. I realize that by not following these rules, I will be prohibited from working in the Oladokun Laboratory.

Signature: _____ **Date:** _____

CHAPTER

02

General Laboratory Practices

Comprehensive In Vitro Laboratory Assay Handbook — General Laboratory Practices, Experimental Design and Universal Sample Preparation

A concise laboratory SOP manual for biological, microbiological, biomaterial, nanomaterial and environmental test samples.

Sections: 2.1 Safety & biosafety · 2.2 PPE · 2.3 Organization · 2.4 Cleaning & decontamination · 2.5 Waste · 2.6 Equipment · 2.7 Documentation · 2.8 Experimental planning · 2.9 Sample receipt & preparation · 2.10 Reagents & buffers · 2.11 Labeling, storage & transport · 2.12 Controls · 2.13 Replicates, randomization & blinding · 2.14 Data recording & backup · 2.15 Statistical planning · 2.16 Data integrity & QA

2.1 Laboratory Safety and Biosafety

Purpose — To ensure laboratory work is performed safely and according to institutional requirements.

Materials / records required

- Approved laboratory SOPs
- Required PPE
- Spill kit
- Disinfectant
- Waste containers
- Safety data sheets

Procedure

- 1 Review the protocol and relevant hazard information before beginning work.
- 2 Identify biological, chemical, physical and equipment hazards.
- 3 Confirm the required containment level and approvals are in place.
- 4 Prepare disinfectant, waste containers and spill-response materials before starting.
- 5 Keep emergency exits, eyewash stations and safety equipment accessible.
- 6 Report spills, exposures, injuries and unsafe conditions immediately.
- 7 Decontaminate the work area after completing the procedure.

CRITICAL

- Treat unknown biological samples as potentially hazardous.
- Do not perform unapproved procedures with infectious agents or hazardous chemicals.
- Follow the institution's exposure-response procedure after an incident.

Checklist

TASK	COMPLETED
Protocol reviewed	☐
Hazards identified	☐
Containment approved	☐
Spill materials ready	☐
Work area decontaminated	☐

2.2 Personal Protective Equipment

Purpose — To define the minimum PPE required for routine laboratory work.

Materials required

- Lab coat or gown
- Disposable gloves
- Safety glasses
- Closed-toe shoes
- Face protection when required

Procedure

- 1 Inspect PPE before use.
- 2 Put on the lab coat or gown before entering the work area.
- 3 Wear gloves appropriate for the materials handled.

- 4 Wear eye and face protection when splash, aerosol or impact hazards are present.
- 5 Change contaminated or damaged gloves immediately.
- 6 Remove gloves before touching phones, door handles or personal items.
- 7 Remove PPE in the correct order and wash hands before leaving.

CRITICAL

- PPE selection must match the hazard.
- Gloves do not replace hand washing.

Checklist

TASK	COMPLETED
Lab coat worn	☐
Gloves appropriate	☐
Eye protection available	☐
Closed-toe shoes worn	☐
Hands washed	☐

2.3 Laboratory Organization

Purpose — To maintain a clean, efficient and traceable laboratory workspace.

Materials / records required

- Storage labels
- Inventory records
- Designated clean and dirty areas
- Secondary containment

Procedure

- 1 Keep benches free of unnecessary materials.

- 2 Separate clean preparation areas from contaminated work areas.
- 3 Label shelves, refrigerators, freezers and cabinets clearly.
- 4 Store incompatible chemicals separately.
- 5 Keep frequently used materials in designated locations.
- 6 Maintain an updated inventory of samples, reagents and consumables.
- 7 Return equipment and supplies to assigned locations after use.

CRITICAL

- Do not store samples or reagents in unlabeled containers.
- Do not block ventilation openings or emergency equipment.

Checklist

TASK	COMPLETED
Bench organized	☐
Storage areas labeled	☐
Inventory updated	☐
Incompatible chemicals separated	☐

2.4 Cleaning and Decontamination

Purpose — To standardize routine cleaning and contamination-control procedures.

Materials required

- Approved detergent
- Institution-approved disinfectant
- Absorbent materials
- PPE
- Waste bags

Procedure

- 1 Remove unnecessary items from the work surface.
- 2 Clean visible dirt using an appropriate detergent.
- 3 Apply the approved disinfectant at the required concentration.
- 4 Keep the surface wet for the validated contact time.
- 5 Wipe from cleaner areas toward more contaminated areas.
- 6 Dispose of used materials in the appropriate waste stream.
- 7 Clean reusable equipment according to manufacturer instructions.
- 8 Document decontamination when required.

CRITICAL

- Do not mix disinfectants unless specifically permitted.
- Confirm material compatibility before disinfecting sensitive equipment.

Checklist

TASK	COMPLETED
Surface cleaned	☐
Correct disinfectant used	☐
Contact time completed	☐
Waste discarded correctly	☐

2.5 Waste Management

Purpose — To ensure correct segregation and disposal of laboratory waste.

Materials required

- Biohazard bags
- Sharps containers

- Chemical-waste containers
- Broken-glass container
- Labels

Procedure

- 1 Identify the waste type before beginning the experiment.
- 2 Place biological waste in the designated biohazard container.
- 3 Place needles, blades and contaminated sharps in approved sharps containers.
- 4 Place chemical waste in a compatible, labeled container.
- 5 Place uncontaminated broken glass in the designated glass container.
- 6 Do not overfill any waste container.
- 7 Close and label waste containers before removal.
- 8 Follow institutional collection and disposal procedures.

CRITICAL

- Never recap needles unless an approved procedure requires it.
- Do not pour chemicals into the sink unless specifically authorized.

Checklist

TASK	COMPLETED
Waste streams identified	☐
Sharps container available	☐
Chemical waste labeled	☐
Containers not overfilled	☐

2.6 Equipment Inspection and Maintenance

Purpose — To ensure equipment performs reliably and safely.

Materials / records required

- Equipment log
- Calibration records
- Manufacturer manuals
- Cleaning materials

Procedure

- 1 Inspect equipment before use for damage, contamination or missing parts.
- 2 Confirm calibration or verification is current.
- 3 Record equipment identification and settings used.
- 4 Operate equipment according to manufacturer and laboratory SOP.
- 5 Clean equipment after use.
- 6 Document errors, unusual sounds, temperature deviations or performance problems.
- 7 Remove defective equipment from service and label it clearly.
- 8 Record maintenance and return-to-service approval.

CRITICAL

Do not use equipment with expired calibration or unresolved faults.

Checklist

TASK	COMPLETED
Inspection completed	☐
Calibration current	☐
Settings recorded	☐
Equipment cleaned	☐
Faults documented	☐

2.7 Laboratory Documentation and Notebooks

Purpose — To create complete, traceable and reproducible laboratory records.

Materials required

- Bound notebook or approved electronic system
- Permanent ink
- Sample labels
- Protocol version

Procedure

- 1 Record the date, experiment title and objective.
- 2 Record sample IDs, treatment groups and replicate numbers.
- 3 Record reagent names, manufacturers, lot numbers and expiry dates.
- 4 Record equipment IDs and settings.
- 5 Record all volumes, concentrations, times and temperatures.
- 6 Record observations and unexpected events immediately.
- 7 Document protocol deviations and corrective actions.
- 8 Sign or electronically authenticate the record.

CRITICAL

- Do not erase original entries.
- Correct paper records with a single line, initials, date and explanation.

Checklist

TASK	COMPLETED
Objective recorded	☐
Sample IDs recorded	☐
Reagent lots recorded	☐

TASK	COMPLETED
Instrument settings recorded	☐
Deviations documented	☐

2.8 Experimental Planning

Purpose — To define the study design before laboratory work begins.

Materials required

- Written protocol
- Plate map
- Sample list
- Statistical plan
- Calculation sheet

Procedure

- 1 Define the research question and primary outcome.
- 2 Specify the test sample, concentrations and exposure times.
- 3 Define untreated, vehicle, positive and negative controls.
- 4 Specify biological and technical replicate numbers.
- 5 Prepare a plate, tube or sample layout.
- 6 Define inclusion, exclusion and repeat criteria.
- 7 Select the planned statistical analysis.
- 8 Review material and equipment availability.
- 9 Approve the final experimental plan before starting.

CRITICAL

Avoid changing the primary outcome after seeing results unless the change is documented as exploratory.

Checklist

TASK	COMPLETED
Objective defined	☐
Controls defined	☐
Replicates assigned	☐
Layout prepared	☐
Statistics planned	☐

2.9 Universal Sample Receipt and Preparation

Purpose — To prepare different sample types consistently before analysis.

Materials required

- Sample-receipt log
- Calibrated balance
- Pipettes
- Sterile or clean tubes
- Appropriate diluent or buffer
- Labels

Procedure

- 1 Inspect the sample container and transport condition on receipt.
- 2 Assign a unique sample identification code.
- 3 Record the source, date, quantity and storage requirement.
- 4 Quarantine samples with leakage, unclear identity or unacceptable transport conditions.
- 5 Mix liquids by gentle inversion, pipetting or vortexing as appropriate.
- 6 Homogenize solid or semi-solid samples using a validated method.
- 7 Prepare a stock concentration using the selected solvent, vehicle or buffer.

- 8 Prepare working concentrations using recorded calculations.
- 9 Aliquot samples when repeated freeze–thaw cycles may affect stability.
- 10 Record all preparation details and observations.

CRITICAL

- Use low-binding containers for samples prone to adsorption.
- Particulate samples must be mixed consistently before each transfer.

Checklist

TASK	COMPLETED
Sample inspected	☐
Unique ID assigned	☐
Storage verified	☐
Stock prepared	☐
Working dilutions labeled	☐

2.10 Reagent and Buffer Preparation

Purpose — To standardize reagent calculation, preparation, labeling and storage.

Materials required

- Analytical balance
- Calibrated pipettes
- Volumetric glassware
- Deionized water
- pH meter
- Sterile filter when required

Procedure

- 1 Review the formula and required final concentration.
- 2 Verify the calculation independently.
- 3 Record reagent names, grades, lot numbers and expiry dates.
- 4 Weigh or measure each component accurately.
- 5 Dissolve components in approximately **80–90%** of the final volume.
- 6 Adjust pH at the required temperature when applicable.
- 7 Bring the solution to the final volume.
- 8 Sterilize by autoclaving or membrane filtration when required.
- 9 Label with name, concentration, pH, date, expiry and preparer's initials.
- 10 Store under the validated condition.

CRITICAL

- Use the same buffer as the blank where buffer background may affect an assay.
- Do not use reagents showing contamination, precipitation or unexpected discoloration.

Checklist

TASK	COMPLETED
Calculation checked	☐
Components recorded	☐
pH adjusted	☐
Final volume confirmed	☐
Label complete	☐

2.11 Sample Labeling, Storage and Transport

Purpose — To maintain sample identity and integrity throughout the study.

Materials required

- Water-resistant labels
- Inventory sheet
- Temperature log
- Secondary transport container

Procedure

- 1 Label each container with the unique sample ID.
- 2 Add sample name, concentration, date, initials and storage condition where space permits.
- 3 Record the exact storage location in the inventory.
- 4 Store samples at the validated temperature and light condition.
- 5 Use aliquots to limit freeze–thaw cycles.
- 6 Transport samples in leak-proof primary and secondary containers.
- 7 Maintain required temperature control during transport.
- 8 Record chain of custody when required.
- 9 Document thawing, transfer and disposal.

CRITICAL

- Do not rely on cap color as the only identifier.
- Quarantine mislabeled or temperature-compromised samples.

Checklist

TASK	COMPLETED
Label complete	☐
Inventory location recorded	☐
Temperature verified	☐
Transport container suitable	☐

2.12 Experimental Controls

Purpose — To ensure assay results can be interpreted correctly.

Materials required

- Plate or tube layout
- Vehicle
- Positive-control material
- Blank matrix

Procedure

- 1 Include an untreated control when applicable.
- 2 Include a vehicle control matching the highest final vehicle concentration.
- 3 Include a positive control that confirms assay responsiveness.
- 4 Include a negative control when a non-response is required.
- 5 Include medium, reagent or instrument blanks.
- 6 Include sample-only blanks when color, turbidity, fluorescence or particles may interfere.
- 7 Apply controls using the same volume, timing and handling as test groups.
- 8 Define control acceptance criteria before analysis.

CRITICAL

A failed control invalidates interpretation of the affected experiment.

Checklist

TASK	COMPLETED
Untreated control included	☐
Vehicle matched	☐
Positive control included	☐
Blanks included	☐

TASK	COMPLETED
Acceptance criteria defined	☐

2.13 Replicates, Randomization and Blinding

Purpose — To reduce bias and improve reproducibility.

Materials required

- Randomization list
- Plate map
- Coding sheet

Procedure

- 1 Define biological and technical replicates before the experiment.
- 2 Use independent biological replicates whenever possible.
- 3 Randomly assign samples or treatments to positions when position effects are possible.
- 4 Balance treatment groups across plates, days or operators.
- 5 Code samples for blinded acquisition or analysis when practical.
- 6 Keep the coding key separate until predefined analysis is complete.
- 7 Record the randomization and blinding method.

CRITICAL

- Technical replicates do not replace independent biological replicates.
- Avoid placing all controls or treatments on one plate edge.

Checklist

TASK	COMPLETED
Biological replicates defined	☐
Technical replicates defined	☐

TASK	COMPLETED
Randomization completed	☐
Blinding recorded	☐

2.14 Data Recording and Backup

Purpose — To preserve original data and maintain traceability.

Materials required

- Primary storage location
- Backup location
- File-naming convention
- Metadata sheet

Procedure

- 1 Save raw instrument files without modification.
- 2 Use a consistent file name containing date, experiment and sample identifiers.
- 3 Export processed data to a separate file.
- 4 Document every transformation, exclusion and normalization.
- 5 Back up raw and processed data to an approved second location.
- 6 Restrict access according to institutional policy.
- 7 Retain versions rather than overwriting final files.
- 8 Verify that backups can be opened.

CRITICAL

- Screenshots do not replace raw instrument files.
- Do not delete outliers without predefined justification.

Checklist

TASK	COMPLETED
Raw files saved	☐
File names standardized	☐
Processing documented	☐
Backup verified	☐

2.15 Statistical Planning

Purpose — To select appropriate analyses before data collection.

Materials required

- Statistical software
- Experimental-design summary
- Primary endpoint definition

Procedure

- 1 Define the experimental unit.
- 2 Define the primary endpoint and secondary endpoints.
- 3 Specify whether samples are paired or independent.
- 4 Estimate sample size when required.
- 5 Choose the planned statistical test.
- 6 Define the significance level and multiple-comparison method.
- 7 Define how missing values and excluded data will be handled.
- 8 Check assumptions before applying parametric tests.
- 9 Report effect sizes and confidence intervals when possible.

CRITICAL

Do not treat technical replicate wells as independent biological observations.

Checklist

TASK	COMPLETED
Experimental unit defined	☐
Primary endpoint defined	☐
Test selected	☐
Multiple comparisons planned	☐

2.16 Data Integrity and Quality Assurance

Purpose — To ensure laboratory data are accurate, complete, attributable and reproducible.

Materials required

- Quality-control records
- Calibration logs
- Deviation form
- Corrective-action record

Procedure

- 1 Confirm controls meet predefined acceptance criteria.
- 2 Verify sample identity and reagent validity.
- 3 Confirm equipment calibration and performance.
- 4 Review raw data for completeness and transcription errors.
- 5 Document deviations, failed controls and repeated experiments.
- 6 Investigate unexpected results using a structured root-cause approach.
- 7 Implement corrective and preventive actions when needed.

- 8 Conduct periodic review of notebooks, inventories and calibration logs.
- 9 Retain records according to institutional policy.

CRITICAL

Quality assurance should be performed throughout the workflow, not only after analysis.

Checklist

TASK	COMPLETED
Controls passed	☐
Calibration verified	☐
Raw data reviewed	☐
Deviations documented	☐
Corrective actions completed	☐

CHAPTER

03

Laboratory Equipment & Instrumentation

*Correct operation, routine settings and safety for the core bench instruments of
the Oladokun Laboratory*

Every student must read the manufacturer's manual and **demonstrate proficiency to the PI** before independently operating any instrument (see §1.2). This chapter lists the core bench equipment of the Oladokun Laboratory, grouped by workflow, with the correct use, strengths, watch-outs and capacity of each. It supplements — it does not replace — the manufacturer's operating manual or the laboratory Safety Manual.

INSTRUMENTS AT A GLANCE

§	GROUP	INSTRUMENTS
3.1	Biosafety & Containment	Biosafety cabinet · anaerobic chamber · fire & gas safety
3.2	Incubation & Culture	CO ₂ , convection & microbiology incubators · shaking water bath
3.3	Microscopy & Imaging	Inverted & phase-contrast microscopes · gel-doc enclosure
3.4	Sample Prep & Handling	Balance · centrifuges · stirrer · baths · chiller · vortex · bead mill · oven · pipettes
3.5	Nucleic-Acid & Molecular	QuantStudio 5 qPCR · MiniAmp & GeneAmp cyclers · spec/fluorometers · power supply
3.6	Cold Storage	4 °C fridge · -20 °C freezer · -80 °C ULT freezer
3.7	Facilities & Support	UPS · label & document printers

3.1 Biosafety & Containment

Biological Safety Cabinet — Labconco Purifier® Logic+ Class II (A2)

Protects the operator, sample and environment with HEPA-filtered downflow and negative-pressure inflow.

- **Use** — All BSL-1 and BSL-2 manipulations (e.g. Salmonella, Campylobacter); work at least **4 in** inside the grille without blocking the air curtain.
- **Strengths** — “Smart-Start” auto-starts and idles the blower with the sash; quiet ECM motor (< **60 dB**); line-of-sight touchscreen airflow monitor.

- **Watch-outs** — Requires routine certification; alarms trigger at **20%** and **10%** remaining HEPA-filter life.
- **Capacity** — 3' / 4' / 5' / 6' widths; maximum safe sash opening **22.5 in.**

Anaerobic Chamber — Coy Vinyl Anaerobic Chamber

Maintains a strict oxygen-free (0–5 ppm O₂) atmosphere for obligate anaerobes.

- **Use** — Culturing, manipulating and incubating anaerobes; a palladium catalyst scrubs residual oxygen.
- **Strengths** — Gloveless sleeve-vacuum system gives bare-handed dexterity; generous footprint fits an internal incubator and vacuum airlock.
- **Watch-outs** — Needs a continuous **5%** hydrogen gas mix and routine high-temperature baking of the catalyst; the vinyl barrier can puncture.

Fire Safety & Compressed Gas — ABC Extinguisher • Gas Cylinders • Dry Ice

Emergency fire suppression and safe handling of compressed and cryogenic gases.

CRITICAL

Always secure high-pressure gas cylinders to a wall or bench. When handling dry ice (solid CO₂) for transport, ensure adequate ventilation — sublimating CO₂ gas is a genuine asphyxiation hazard.

3.2 Incubation & Culture

CO₂ Incubator — NuAire US AutoFlow (NU-4500)

Water-jacketed automatic CO₂ incubator holding stable physiological conditions.

- **Use** — Mammalian cell culture at ~**5% CO₂** to keep bicarbonate-buffered media at physiological pH.
- **Strengths** — The water jacket buffers strongly against room-temperature swings and power loss.
- **Capacity** — 4 stainless-steel removable shelves and a full-size humidity water pan.

CRITICAL

Use only single-distilled or de-ionised water ($\leq 5 \text{ M}\Omega$, no purer). No chlorinated or halogen materials in the jacket or chamber. Calibrate the CO₂ sensor periodically with a Fyrite device.

Mechanical Convection Incubator — Precision™

Forced-air incubation giving uniform temperature across all shelves.

- **Use** — Culturing bacteria or yeast on agar plates or in liquid media.
- **Strengths** — A built-in fan drives convection for highly uniform temperature distribution.
- **Watch-outs** — Forced air dries agar plates faster than gravity models — wrap or seal plates.
- **Capacity** — Multiple adjustable shelves; hundreds of standard petri dishes.

Microbiology Incubator — Fisher Scientific Isotemp®

Gentle gravity-convection incubation designed for microbiological samples.

- **Use** — Growth where draft-free air is preferred to prevent desiccation.
- **Strengths** — Robust design, often with a secondary inner glass door for viewing without heat loss.
- **Watch-outs** — Temperature recovery after the door opens is slower than mechanical-convection units.
- **Capacity** — Typically 60–150 L with configurable shelving.

Shaking Water Bath — Lab-Line Instruments Microprocessor Shaker (3527 series)

Benchtop incubator combined with an orbital shaking mechanism.

- **Use** — Bacterial culture and work requiring simultaneous temperature control and aeration.
- **Strengths** — Clear hood for viewing; optional cooling coil for below-ambient control.
- **Watch-outs** — The blower fan must be on before heating; safe maximum speed may be well below the 400 rpm dial depending on load.
- **Capacity** — 18" × 18" platform; flasks and beakers up to 2 L.

3.3 Microscopy & Imaging

Inverted Microscope — Motic AE20 / AE21

Inverted biological microscope optimised for specimens in culture vessels.

- **Use** — Brightfield and phase-contrast viewing of live, unstained cells and morphology checks.
- **Strengths** — Independent left-eyepiece diopter reduces focus loss between magnifications; dedicated “Ph” objectives simplify phase-contrast.
- **Watch-outs** — The focus mechanism is easily damaged by rotating the knobs against each other or past their limits.

Phase-Contrast Microscope — Parco

Biological microscope optimised for phase-contrast, high-magnification observation.

- **Use** — Live, unstained specimens and cultures where natural contrast is low.
- **Strengths** — Fully coated optics for sharp images; independent diopter adjustment.
- **Watch-outs** — Manual focus gears are damaged by forcing the tension knobs past their limits.

Gel Documentation Enclosure — Accuris SmartDoc™

Imaging enclosure that documents electrophoresis gels with a smartphone camera.

- **Use** — Capturing DNA, RNA or protein bands over a blue-light or UV transilluminator.
- **Strengths** — Economical; built-in photo filters; sits directly on standard transilluminators.
- **Watch-outs** — Resolution and raw-data quality are limited by the phone camera, not a scientific CCD sensor.

3.4 Sample Preparation & Reagent Handling

Analytical & Precision Balance — VWR Advanced Series (A-Series)

High-precision sample weighing and formulation.

- **Use** — Weighing, part counting, check weighing, dosing and density determination.
- **Strengths** — Automatic internal calibration on time or ambient change ($\geq 3\text{ }^{\circ}\text{C}$); GLP printouts; tool-free draft shield.
- **Capacity** — Analytical models (0.1 mg readability) to 520 g; precision variants to 10.1 kg.

Centrifuges — Fisherbrand accuSpin 21R • Thermo Sorvall ST 16

Density separation — pelleting microbial cells, precipitating nucleic acids and separating slurries.

- **Use** — Refrigerated options for temperature-sensitive cell and vesicle work.
- **Capacity** — Swing-bucket and fixed-angle rotors; refrigerated to 4 °C.

CRITICAL

Balance every rotor position against its opposite by MASS, not volume — an unbalanced spin is a severe safety hazard. Never clean rotors with bleach (it causes pitting and stress corrosion); use pH-neutral detergent and never fully submerge a rotor.

Digital Hotplate Stirrer — Corning PC-420D

Combined magnetic stirring and heating with digital readouts.

- **Use** — Preparing media, dissolving reagents and mixing viscous or semi-solid solutions.
- **Strengths** — Holds 5–550 °C; closed-loop stirring 60–1150 rpm; optional external probe for in-vessel control.
- **Capacity** — 5 × 7 in Pyroceram® glass-ceramic top.

Water Bath — Lab Companion BW-05G

Precision constant-temperature liquid heating.

- **Use** — Thawing frozen cells, heat-shock transformations and warming media.
- **Strengths** — PID control gives excellent stability and uniformity to 100 °C.
- **Watch-outs** — Monitor the water level or the heating elements are exposed and burn out.
- **Capacity** — Holds up to 4 L.

Recirculating Chiller — VWR Circulator / Chiller

Temperature-controlled fluid circulation to external equipment.

- **Use** — Cooling electrophoresis units, jacketed bioreactors and condensers.
- **Strengths** — Closed-loop, continuous sub-ambient cooling for heat-sensitive samples.
- **Watch-outs** — Needs an antifreeze/water mix (e.g. ethylene glycol), not pure water, for sub-zero operation.

- **Capacity** — 6–10 L reservoirs with cooling well below 0 °C.

Vortex Mixer — IKA MS 3

Turbulent oscillation mixing.

- **Use** — Mixing reagents or disrupting loose cell clumps.
- **Watch-outs** — Prolonged high-intensity vortexing shears high-molecular-weight DNA — avoid before long-read sequencing (Nanopore, PacBio).

Bead Mill Homogenizer — Fisherbrand™ Bead Mill 24

High-impact physical disruption of samples by bead beating.

- **Use** — Lysing tough tissues, microbes and spores before nucleic-acid or protein extraction.
- **Strengths** — Thorough homogenates and higher yields; stores up to 99 programs.
- **Watch-outs** — High-intensity agitation generates heat — process temperature-sensitive samples in short bursts.
- **Capacity** — Up to 24 × 2.0 mL or 0.5 mL tubes.

Drying / Sterilizing Oven — Constant Temperature Oven (KTD-8000)

Dry-heat drying, baking and sterilizing of glassware and solids.

- **Use** — Heat sterilization of glassware and prolonged sample drying.
- **Strengths** — PID control with programmable timing and temperature-deviation alarms.
- **Watch-outs** — Not for incubating biological samples — too hot and no humidity control.

Micropipettes — Air-Displacement Hand Pipettes (P2–P1000)

Precise transfer of 0.2–1000 µL; choose the smallest pipette whose range covers the target volume.

- **Use** — Aspirate to the first stop, dispense to the second; pre-wet a fresh tip; keep upright with a filled tip; send for calibration annually.
- **Capacity** — P2 0.2–2 µL · P10 1–10 µL · P20 2–20 µL · P100/P200 20–200 µL · P1000 100–1000 µL.

Serological Pipettes & Controller

Sterile transfer of 1–50 mL volumes with a motorised or bulb controller.

- **Use** — Match pipette size to volume; seat gently into the controller (never force); keep the tip below the surface; use a fresh pipette per reagent.
- **Capacity** — 1, 2, 5, 10, 25 and 50 mL graduated pipettes.

3.5 Nucleic-Acid & Molecular Analysis

Real-Time PCR System — Applied Biosystems QuantStudio 5

Fluorescence-based PCR for quantitative and qualitative nucleic-acid detection and melt-curve analysis.

- **Use** — Standard-curve and relative ($\Delta\Delta C_t$) quantification and genotyping; assays set up from an Experiment Template (EDT) on the touchscreen or desktop software.
- **Strengths** — 6 programmable VeriFlex zones for accurate block temperatures; Security, Audit & e-Signature (SAE) mode; Thermo Fisher Connect cloud monitoring.
- **Watch-outs** — With SAE security enabled, runs cannot start from the touchscreen — they must be launched from the desktop software.
- **Capacity** — 96-well (0.2 / 0.1 mL) or 384-well block; 5–6 optical channels.

Thermal Cycler — Applied Biosystems MiniAmp™

Automated temperature cycling for PCR.

- **Use** — Amplifying target DNA for cloning, sequencing or genetic analysis.
- **Strengths** — Very compact (7.5 in wide); 5-in touchscreen; cloud-enabled via Thermo Fisher Connect.
- **Watch-outs** — The standard model has a basic isothermal block — gradient optimisation only on the “Plus” model with VeriFlex zones.
- **Capacity** — One 96-well (0.2 mL) plate or individual tubes.

Legacy Thermal Cycler — PerkinElmer GeneAmp 2400

Basic early-generation temperature cycling for PCR.

- **Use** — Standard endpoint PCR amplification.
- **Strengths** — Durable and proven for routine amplification.

- **Watch-outs** — Legacy, discontinued system — no gradient block, touchscreen or USB transfer.
- **Capacity** — 24-well block for older **0.2 mL** tubes.

Spectrophotometers & Fluorometers — Nanbei UV-Vis · BioTek Synergy H1 · Tecan SpectraFluor

Optical quantification of density and nucleic acids.

- **Use** — Spectrophotometry reads OD₆₀₀ to estimate bacterial density; fluorometry quantifies DNA/RNA accurately with target-specific dyes.
- **Watch-outs** — Absorbance is a crude proxy — it cannot distinguish intact DNA, degraded RNA or free nucleotides and reads non-linearly above **OD 1.0**. Rely on the fluorometer for sequencing workflows.

Electrophoresis Power Supply — Bio-Rad PowerPac™

Provides constant voltage, current or power for electrophoretic separations.

- **Use** — Running vertical protein gels and horizontal agarose DNA gels.
- **Strengths** — Fully programmable with automatic crossover and power-failure detection that resumes runs.
- **Watch-outs** — The Basic model's lower maximum output limits high-demand Western blotting — use HV/HC variants there.
- **Capacity** — Powers up to 4 electrophoresis cells simultaneously.

3.6 Cold Storage & Sample Preservation

Refrigerator (4 °C)

Short-term storage of general reagents, buffers and active culture plates.

- **Use** — Temporarily slowing microbial growth and preserving samples for imminent assays.
- **Watch-outs** — Enzymes retain activity only briefly at **4 °C**; not for long-term storage of proteins, RNA or delicate specimens.

Freezer (−20 °C)

General freezing of reagents, DNA extracts and master mixes.

- **Use** — Robust nucleic acids (stable DNA) and master mixes.

CRITICAL

Prefer MANUAL-defrost units over “frost-free” models for sensitive samples — frost-free freezers run periodic internal heating cycles that subject samples to repeated, damaging freeze–thaw events.

Ultra-Low Temperature Freezer (–80 °C) — Eppendorf CryoCube® F570n • Fisher Isotemp ULT

Extreme cold storage for long-term biobanking — glycerol stocks, unstable RNA and archived extracts.

- **Use** — Aliquot samples into single-use volumes before freezing; use inventory racks and barcodes; keep the door open for the minimum time.
- **Strengths** — 130 mm walls (polyurethane foam + vacuum panels) for long warm-up in a power failure; vent ports equalise pressure for quick re-opening; optional LN₂ or CO₂ backup.
- **Watch-outs** — Generating extreme cold exhausts large heat loads — units need adequate HVAC or they overheat the room (water-cooled variants avoid this).
- **Capacity** — CryoCube F570n: **570 L** (20.1 ft³), up to 25 racks / 400 boxes / ~40,000 × **1.5 mL** vessels; Isotemp range **85–815 L**.

3.7 Facilities & Support

Uninterruptible Power Supply — Tripp Lite OmniSmart

Line-interactive voltage regulation and emergency battery backup.

- **Use** — Safeguarding thermal cyclers, sequencers and sensitive equipment from outages and surges.
- **Strengths** — Automatically corrects brownouts and overvoltages without draining the battery.
- **Watch-outs** — Bridges brief outages or allows a safe shutdown only — it cannot sustain heaters, incubators or ULT freezers.
- **Capacity** — 500–1500 VA; 6–10 protected outlets.

Label & Document Printers — KATASYMBOL T50M Pro • Canon PIXMA TS3720

Thermal label printing and standard document / photo printing.

- **Use** — T50M Pro — durable, cold-resistant labels for tubes and cryo-boxes; TS3720 — protocols, gel images and reports.
- **Strengths** — T50M Pro is portable, inkless Bluetooth; the Canon is low-cost wireless.
- **Watch-outs** — Thermal labels blacken in autoclave heat; inkjet prints run and blur with water or common solvents.

NOTE

Log every use, fault and maintenance action per §2.6 and complete the daily equipment walk-through in §1.4. Remove any instrument with an unresolved fault or expired calibration from service and label it clearly.

CHAPTER

04

Laboratory Protocols

Primary cell culture, cell counting, extracellular-vesicle isolation, protein quantification and in vitro assays

PROTOCOL 4.1

Primary Intestinal Cell Isolation & Culture

Modified from Athman et al. (2005), Guo et al. (2015), and Zaffuto et al. (2008)

4.1.1 Materials and Reagents

ENZYMATIC DIGESTION SOLUTION — 25 mL PER SAMPLE

COMPONENT	STOCK	AMOUNT TO ADD	FINAL CONCENTRATION
Collagenase Type I	Powder	25 mg	1 mg/mL
Hyaluronidase	Powder	2.5 mg	0.1 mg/mL
Dispase	Powder or liquid	25 Units	1 U/mL
HBSS (pH 7.4)	Sterile	Up to 25 mL	Base solution

Hanks' Balanced Salt Solution or PBS may serve as the base.

Preparation

- Add HBSS to a sterile 50 mL tube (~20 mL initially)
- Add Collagenase Type I (25 mg)
- Add Hyaluronidase (2.5 mg)
- Add Dispase (25 Units, or 25 µL)
- Mix gently until dissolved
- Adjust volume to exactly 25 mL with HBSS
- Filter through 0.22 µm filter for sterility
- **Prepare fresh before use**

DMEM-W (WASH MEDIUM) — NO SORBITOL AVAILABLE

For 100 mL:

- DMEM base medium — 97.5 mL
- FBS — 2.5 mL (2.5%)
- **Total:** 100 mL

Store at 4 °C; warm to 37 °C before use.

COMPLETE CULTURE MEDIUM

For 100 mL:

- DMEM base medium — 82 mL
- FBS — 15 mL (15%)
- PSG — 1 mL (1%)
- ITS — 1 mL (1%)
- NEAA — 1 mL (1%)
- **Total:** 100 mL

Store at 4 °C; warm to 37 °C before use.

4.1.2 Step 1 — Tissue Preparation

- Collect fresh chicken intestinal tissue immediately after sacrifice
- Wash tissue **3 times** with sterile PBS containing **1%** Penicillin-Streptomycin
- Open intestine longitudinally using sterile scissors
- Gently scrape off excess mucus with a sterile glass slide or scalpel blade
- Cut tissue into small **2–3 mm²** pieces using a sharp sterile scalpel blade
- Keep tissue pieces in ice-cold sterile PBS until ready for digestion

4.1.3 Step 2 — Enzymatic Digestion

- Transfer tissue pieces into a sterile **50 mL** centrifuge tube containing **25 mL** pre-warmed (**37 °C**) enzymatic digestion solution
- Place tube in an orbital shaker incubator
- Incubate with shaking:
 - **Temperature:** 37 °C
 - **Speed:** 130 rpm
 - **Duration:** 30–60 minutes
- Check digestion progress every **15 minutes** by observing tissue dissociation
- Stop digestion when tissue is well-dissociated by adding **5 mL ice-cold FBS**, or by transferring the tube to ice

4.1.4 Step 3 — Cell Collection (First Centrifugation)

- Centrifuge at **300 × g for 5 min** at room temperature
- Carefully aspirate and discard supernatant (enzymes, debris, dead cells)
- Leave approximately **1 mL** of liquid to avoid disturbing the pellet

4.1.5 Step 4 — Cell Washing

- Add **10 mL sterile PBS** or HBSS to resuspend the pellet
- Pipette gently up and down, or roll on the palm, to disperse cells — avoid bubbles
- Centrifuge at **300 × g for 5 min** at room temperature
- Carefully aspirate supernatant
- Repeat washing **2 more times** (3 washes total)
- After the final wash, aspirate supernatant carefully

4.1.6 Step 5 — Differential Centrifugation (Cell Purification)

- Resuspend cell pellet in **10–15 mL DMEM-W** (DMEM + **2.5%** FBS)
- **First differential spin:**
 - Centrifuge at **50 × g for 3 min** at room temperature — removes large debris and tissue fragments
 - **Carefully transfer supernatant** (contains single cells) to a new sterile **50 mL** tube
 - Leave the pellet behind (debris)
- **Second spin:**
 - Centrifuge collected supernatant at **300 × g for 5 min** — pellets viable cells
 - Aspirate supernatant and resuspend pellet in **10 mL fresh DMEM-W**
- **Repeat differential centrifugation 3–4 times** until the supernatant after the **50 × g** spin is clear
- Centrifuge the clear supernatant at **300 × g**, then aspirate carefully

4.1.7 Step 6 — Final Cell Resuspension

- Resuspend the final cell pellet in **5–10 mL complete culture medium:**
 - DMEM + **15%** FBS + **1%** PSG + **1%** ITS + **1%** NEAA
- Mix gently by pipetting up and down

4.1.8 Step 7 — Cell Counting and Viability Assessment

- Take **10 µL** of cell suspension
- Mix with **10 µL of 0.4% Trypan Blue**
- Load **10 µL** onto a hemocytometer
- Count cells in 4 corner squares under the microscope (10× objective)

Calculations

CALCULATION

Cell concentration (cells/mL) = (Average count per square) × 2 (dilution factor) × 10^4

Viability (%) = (Viable / unstained cells ÷ Total cells) × 100

Total cells = Cell concentration (cells/mL) × Total volume (mL)

Acceptable viability should be $\geq 80\%$

4.1.9 Step 8 — Initial Cell Seeding (Optimization)

Prepare cell suspension at 1×10^6 cells/mL. *Example:* if you have 5×10^6 total cells, add 5 mL complete medium.

Seed three T25 flasks at different densities:

FLASK	CELL SUSPENSION	COMPLETE MEDIUM	TOTAL	CELL NUMBER
Flask 1	0.25 mL	4.75 mL	5 mL	2.5×10^5 cells
Flask 2	0.5 mL	4.5 mL	5 mL	5×10^5 cells
Flask 3	1.0 mL	4.0 mL	5 mL	1×10^6 cells

Incubation

- Swirl flasks gently to distribute cells evenly
- Place in incubator at 37°C , 5% CO_2 , humidified atmosphere
- Do not disturb for the first 24 hours** to allow cell attachment

4.1.10 Step 9 — Flask Selection (24 h after seeding)

- Examine all three flasks under phase-contrast microscope
- Assess:

- Cell attachment efficiency
- Cell spreading and morphology (epithelial cobblestone appearance)
- Confluence level
- Minimal debris
- Select the flask with best overall attachment and morphology; discard the others
- Perform first medium change on the selected flask:
 - Aspirate old medium carefully
 - Add **5 mL** fresh pre-warmed complete medium
 - Return to incubator

4.1.11 Step 10 — Cell Maintenance

- **Medium changes:** replace medium every **2–3 days**
 - Aspirate old medium; add **5 mL** fresh pre-warmed complete medium
- **Daily monitoring:** check cells under microscope for
 - Growth and confluence
 - Morphology
 - Contamination
- **Passage cells** when they reach **80–90% confluence** (typically 3–5 days)

4.1.12 Step 11 — Cell Passaging (Subculture)

When cells reach 80–90% confluence:

- 1 Aspirate culture medium
- 2 Wash with **5 mL sterile PBS**, swirl gently, aspirate (repeat once)
- 3 Add **1 mL of 0.25% Trypsin-EDTA** to the T25 flask
- 4 Incubate at **37 °C for 3–5 min**
- 5 Tap flask gently to dislodge cells (check under microscope — cells should be rounded)

- 6 Add **1 mL complete medium** to neutralize trypsin (total **2 mL**)
- 7 Transfer cell suspension to a sterile **15 mL** centrifuge tube
- 8 Centrifuge at **300 × g for 5 min** at room temperature
- 9 Aspirate supernatant carefully
- 10 Resuspend pellet in **5 mL complete medium**
- 11 Count cells using the Trypan Blue method
- 12 Seed new flasks or experimental plates at the desired density

4.1.13 Step 12 — Standard Seeding Densities

FLASK / PLATE	SURFACE AREA	CELLS TO SEED	MEDIUM VOLUME
T25 flask	25 cm ²	2.5 – 5 × 10 ⁶ cells	5 mL
T75 flask	75 cm ²	7.5 – 15 × 10 ⁶ cells	15 mL
24-well plate	1.9 cm ² /well	2 – 4 × 10 ⁵ cells/well	0.5 – 1 mL/well
96-well plate	0.32 cm ² /well	3 – 6 × 10 ⁴ cells/well	100 – 200 µL/well

Target seeding density: 1 – 2 × 10⁵ cells/cm²

4.1.14 Step 13 — Preparation for Experiments

For experiments in 24-well or 96-well plates:

- **Optional pre-coating** (recommended for better attachment):
 - Prepare collagen I solution at **100 µg/mL** in sterile PBS
 - 24-well plate: add **250–300 µL** per well
 - 96-well plate: add **50–100 µL** per well
 - Incubate **1 hour** at **37 °C**
 - Aspirate collagen solution before seeding cells
- **Seed cells** at appropriate density (see table above)

- **Incubate 24 hours** at 37 °C, 5% CO₂ to allow attachment and equilibration
- **Check confluence:** cells should be **60–80%** confluent before starting experiments
- **Begin experiments 24 hours** after plating

4.1.15 Expected Timeline

DAY	ACTIVITY	EXPECTED OBSERVATION
Day 0	Tissue digestion and cell isolation	Cell suspension obtained
Day 0	Seed cells in T25 flasks (3 densities)	Cells in suspension
Day 1	Select best flask, first medium change	Cells attached, 10–30% confluence
Day 2–3	Monitor and maintain	Cells spreading, 40–60% confluence
Day 4–5	Cells reach 80–90% confluence	Ready for passage
Day 5	Passage to experimental plates (if needed)	Cells seeded for experiments
Day 6	Cells equilibrate in experimental plates	60–80% confluence, ready
Day 7	Begin experiments	Experiments started

4.1.16 Quality Control Indicators

After enzymatic digestion

- Tissue well-dissociated with minimal large clumps
- Cell suspension appears cloudy
- Individual cells visible under microscope

After cell counting

- Cell viability $\geq 80\%$ (Trypan Blue exclusion)
- Good yield (typically $1\text{--}5 \times 10^6$ **cells** per gram of tissue)

After 24 hours in culture

- $\geq 70\%$ of cells attached to flask surface
- Epithelial morphology (polygonal, cobblestone)
- Minimal floating debris
- Clear medium (pink color, stable pH)

Before experiments

- Cells **60–80%** confluent (not over-confluent)
- Uniform monolayer formation
- No signs of contamination
- Proper epithelial morphology maintained

PROTOCOL 4.2

Cell Counting, Passaging & Seeding

Purpose — To determine viable cell concentration and seed equal numbers of cells into 96-well and 24-well plates.

4.2.1 Part 1 — Cell Counting (Hemocytometer)

- 1 Mix 10 µL cell suspension with 10 µL trypan blue.
- 2 Load hemocytometer (10 µL).
- 3 Count cells in the four corner squares.
- 4 Calculate concentration:

> **Cells/mL = Average count × 2 × 10,000**

- 1 Viability = (Live ÷ Total) × 100 — **must be > 90%**

4.2.2 Target Seeding Concentrations

PLATE	CELLS PER WELL	WORKING CONCENTRATION
96-well	25,000 cells/well	250,000 cells/mL
24-well	150,000 cells/well	300,000 cells/mL

4.2.3 Worked Example — Stock = 1,200,000 cells/mL

96-WELL PLATE

- Dilution factor $\approx 1,200,000 \div 250,000 \approx 5\times$
- Mix ratio: **1 part cells + 4 parts media**
- Total volume needed = $(96 \text{ wells} \times 100 \text{ } \mu\text{L}) + \sim 10\% \text{ pipetting loss} \approx 11 \text{ mL}$
- Cells volume = $\text{Total} \div \text{Dilution factor} = 2.2 \text{ mL}$
- Media volume = $\text{Total} - \text{Cells} = 8.8 \text{ mL}$
- Seed **100 μL per well**

24-WELL PLATE

- Dilution factor $\approx 1,200,000 \div 300,000 \approx 4\times$
- Mix ratio: **1 part cells + 3 parts media**
- Total volume needed $\approx 13 \text{ mL}$
- Cells: **3.25 mL** · Media: **9.75 mL**
- Seed **500 μL per well**

4.2.4 Fast Mental Shortcut

Purpose — Rapidly determine dilution and prepare seeding suspension without full calculator-based calculations.

MEMORIZE TARGET CONCENTRATIONS

- 96-well: **25,000 cells/well** $\rightarrow$ **250,000 cells/mL**
- 24-well: **150,000 cells/well** $\rightarrow$ **300,000 cells/mL**

STEP 1 — DETERMINE DILUTION FACTOR

Divide stock concentration by target concentration.

Example: $1,200,000 \div 250,000 \approx 5\times$ **dilution**

STEP 2 — CONVERT TO MIXING RATIO

Dilution – 1 = media parts

DILUTION	RATIO (CELLS : MEDIA)
2×	1 : 1
3×	1 : 2
4×	1 : 3
5×	1 : 4

STEP 3 — CALCULATE TOTAL VOLUME

- 96-well: wells × 0.1 mL
- 24-well: wells × 0.5 mL
- Add ~10% extra for pipetting loss

STEP 4 — SPLIT VOLUMES

- Cells volume = Total ÷ dilution factor
- Media volume = Total – cells volume

EXAMPLE (STOCK = 1,200,000 CELLS/ML)

- 96-well: ~11 mL total → 2.2 mL **cells** + 8.8 mL **media**
- 24-well: ~13 mL total → 3.25 mL **cells** + 9.75 mL **media**

4.2.5 Seeding Procedure

- Warm media to 37 °C
- Make a single-cell suspension
- **Remix frequently while pipetting** — cells settle fast
- Pipette to the center of each well
- Gently swirl the plate

- Incubate **undisturbed for 4–6 hours**
- Seeding volume: **96-well** → 100 $\mu\text{L}/\text{well}$ • **24-well** → 500 $\mu\text{L}/\text{well}$

PROTOCOL 4.3

Crude Extracellular Vesicle (EV) Isolation from *Clostridium perfringens*

Polymer-Based Precipitation Method using Total Exosome Isolation Reagent (Thermo Cat. # 4478359)

4.3.1 Materials & Reagents

- *Clostridium perfringens* culture
- Appropriate anaerobic broth medium
- Anaerobic culture system
- Sterile PBS (Ca^{2+} / Mg^{2+} free)
- 0.45 μm syringe filters
- 0.22 μm syringe filters
- Sterile syringes
- Total Exosome Isolation Reagent (Thermo Cat. # 4478359)
- Protease inhibitor cocktail (optional)

4.3.2 Equipment

- Refrigerated centrifuge
- Heat block or water bath
- Ice bucket
- Sterile low-protein-binding tubes

- Pipettes and sterile tips

4.3.3 Procedure

STEP 1 — BACTERIAL CULTURE PREPARATION

- Inoculate *Clostridium perfringens* into anaerobic broth medium
- Incubate anaerobically at **37 °C overnight (16–18 h)**

STEP 2 — REMOVAL OF BACTERIAL CELLS

- Transfer culture into centrifuge tubes
- Centrifuge at **10,000 × g for 15 min at 4 °C**
- Carefully transfer supernatant into new sterile tubes without disturbing the pellet

STEP 3 — FILTRATION OF SUPERNATANT

- Filter supernatant first through a **0.45 µm** filter
- Filter again through a **0.22 µm** filter
- Collect sterile cell-free supernatant

STEP 4 — POLYMER-BASED EV PRECIPITATION

- Add **5 mL Total Exosome Isolation Reagent** to every **10 mL** filtered supernatant
- Mix gently by inversion
- **Do NOT vortex**

STEP 5 — INCUBATION

- Incubate samples at **4 °C overnight (12–16 h)**

STEP 6 — EV PELLETING

- Centrifuge at **10,000 × g for 30 min at 4 °C**
- Carefully discard supernatant
- Pellet may appear small or invisible

STEP 7 — EV RESUSPENSION

- Resuspend pellet in **150 μ L sterile PBS** (per every **5 mL** Isolation Reagent + **10 mL** supernatant)
- Pipette gently up and down until fully resuspended
- **Optional:** resuspend the EV pellet in **150 μ L** sterile PBS containing 1 $\times$ protease inhibitor.
For a 100 $\times$ protease inhibitor stock, mix **1.5 μ L protease inhibitor cocktail with 148.5 μ L sterile PBS**

STEP 8 — HEAT-INACTIVATED EV CONTROL

- Aliquot crude EV preparation into sterile microtubes
- Incubate at **65 $^{\circ}$ C for 30 min**
- Immediately place on ice for **5 min**

STEP 9 — CLARIFICATION

- Centrifuge at **10,000 $\times$ g for 5 min**
- Transfer supernatant into a fresh sterile tube
- Label as: **Heat-inactivated crude EV**

4.3.4 Storage

- **Short-term:** 4 $^{\circ}$ C ($\leq$ 24 h)
- **Long-term:** -80 $^{\circ}$ C
- **Avoid repeated freeze-thaw cycles**

4.3.5 Minimal Quality Control

- Use **BCA assay** for protein quantification
- Plate a small aliquot of the EV preparation on agar to **confirm sterility**

4.3.6 Experimental Controls

- PBS-only control
- Crude EV preparation
- Heat-inactivated crude EV
- Precipitated sterile medium control
- Unprecipitated culture supernatant

PROTOCOL 4.4

Bradford Assay Quantification of Crude EVs

This SOP describes the complete procedure for determining the protein concentration of crude extracellular vesicles using Thermo Scientific Bradford Protein Assay Reagent and Thermo Scientific Albumin Standard (Cat. #23209, 2 mg/mL).

4.4.1 Purpose

To quantify crude EV-associated protein concentration for normalization of antibacterial, MTT, ROS, and cytokine assays.

4.4.2 Materials and Reagents

- Bradford Protein Assay Reagent
- Thermo Albumin Standard Cat. #23209 (2 mg/mL)
- Crude EV samples suspended in PBS
- PBS
- 96-well flat-bottom microplate
- Microcentrifuge tubes
- Single-channel or multichannel pipettes
- Plate reader capable of reading at 595 nm

4.4.3 Preparation of BSA Standards

Prepare 100 µL of each standard as shown.

TUBE	FINAL CONC. (MG/ML)	BSA (2 MG/ML)	PBS
Blank	0	0 μ L	100 μ L
1 S1	125	6.25 μ L	93.75 μ L
2 S2	250	12.5 μ L	87.5 μ L
3 S3	500	25 μ L	75 μ L
4 S4	1000	50 μ L	50 μ L
5 S5	1500	75 μ L	25 μ L
6 S6	2000	100 μ L	0 μ L

4.4.4 EV Sample Preparation

- Thaw EV samples **on ice**
- Mix gently by pipetting — **do not vortex**
- Prepare one **undiluted** sample
- Prepare a **1:5 dilution**: 20 μ L EV + 80 μ L PBS
- Label all tubes clearly

4.4.5 Plate Setup

Run all standards and samples in **triplicate**.

GROUP	REP 1	REP 2	REP 3
Blank	☐	☐	☐
125 μ g/mL	☐	☐	☐
250 μ g/mL	☐	☐	☐
500 μ g/mL	☐	☐	☐

GROUP	REP 1	REP 2	REP 3
1000 $\mu\text{g/mL}$	☐	☐	☐
1500 $\mu\text{g/mL}$	☐	☐	☐
2000 $\mu\text{g/mL}$	☐	☐	☐

For each well:

- 5 μL standard or EV sample
- 200 μL Bradford reagent
- **Total volume = 205 μL per well**

4.4.6 Incubation

Incubate the plate at room temperature for **10 minutes**, protected from direct light.

4.4.7 Absorbance Measurement

Read absorbance at **595 nm** using a microplate reader.

4.4.8 Data Analysis

- Average the triplicate absorbance values for each standard and sample
- Generate a standard curve — concentration versus absorbance
- Determine EV protein concentration from the standard curve equation

4.4.9 Example Calculation

CALCULATION

Standard curve equation: $y = 0.0005x + 0.05$ Measured EV absorbance = **0.65**

$0.65 = 0.0005x + 0.05 \rightarrow x = 1200 \text{ } \mu\text{g/mL}$

If a 1:5 dilution was measured: **Actual EV concentration** = $1200 \times 5 = 6000 \text{ } \mu\text{g/mL} = 6 \text{ mg/mL}$

4.4.10 Determining EV Treatment Volume

CALCULATION

Example: desired final concentration = $10 \text{ } \mu\text{g/mL}$ in a $100 \text{ } \mu\text{L}$ cell culture well.

Protein required = $10 \text{ } \mu\text{g/mL} \times 0.1 \text{ mL} = 1 \text{ } \mu\text{g}$

If EV stock = $1200 \text{ } \mu\text{g/mL}$: Required EV volume = $1 \div 1200 = 0.83 \text{ } \mu\text{L}$

4.4.11 Suggested EV Treatment Range

TREATMENT GROUP	FINAL EV CONCENTRATION
Low	$5 \text{ } \mu\text{g/mL}$
Medium	$10 \text{ } \mu\text{g/mL}$
High	$20 \text{ } \mu\text{g/mL}$

4.4.12 Notes

- Always include a **PBS blank**
- Keep EVs **on ice** during preparation
- Avoid repeated freeze–thaw cycles
- Run standards and samples in **triplicate**

- Bradford measures **total crude EV-associated protein**

PROTOCOL 4.5

In Vitro Protocols for Crude Test Samples from *Clostridium perfringens*

NOTE

N.B. These protocols can be used for any samples — just substitute the EVs with your test samples. This document provides assay volumes and analytical workflows but does not replace approved procedures for culturing or propagating pathogenic bacteria.

4.5.1 Scope and Experimental Logic

Designed for crude test samples ("samples") extracted from *Clostridium perfringens* (CP). Proposed sample doses are **1, 5 and 10** µg/mL, normalized by total sample protein concentration measured by BCA assay.

The workflow has four parts:

PART	FOCUS
A	Sample characterization: BCA, sterility/no-live-CP check, SDS-PAGE, toxin screening, sample-associated DNA PCR, TEM, optional DLS/NTA
B	Antibacterial / anti-virulence testing against CP: growth readout, viable count, live/dead staining, diffusion assays, MIC/MBC, biofilm assay, virulence qRT-PCR, SEM
C	Chicken-cell protection assays: cytotoxicity, viability, LDH, ROS, JC-1, Annexin V/PI, qRT-PCR, immunofluorescence, cytokine ELISA
D	Imaging: TEM for samples; SEM for CP morphology, biofilm architecture and chicken-cell protection

Assays are adapted from Jiang et al. (2014), Nicholas et al. (2017), Luo et al. (2020), Choi et al. (2020) and Mei et al. (2020). Where a step is directly based on one of these studies, the citation is stated in that protocol section.

4.5.2 Biosafety and Sample-Quality Requirements

- Treat live CP, CP-contaminated materials, test sample preparations before sterility confirmation, and CP-challenged cells as **potentially infectious**. Use the containment level, PPE, transport, decontamination and waste procedures approved by the Oladokun Lab and the biosafety office.
- Before applying test samples to chicken cells, perform a **no-live-CP / sterility check** using the laboratory-approved CP sterility method. **Do not use sample lots that show viable CP growth** in cell-culture protection assays.
- For cell protection studies, the cleanest first model is a **sterile CP challenge material** — filtered toxin-containing CP supernatant or a purified toxin/challenge preparation. If live CP challenge is used, use only the approved MOI, media, incubation and neutralization conditions.
- Because test samples may contain soluble toxins, free proteins or nucleic acids, include **vehicle control, mock-sample control and toxin screening** to avoid overinterpreting effects as sample-specific.

4.5.3 Preparation of Sample Working Solutions

Prepare fresh working solutions **on ice immediately before treatment**. Use sterile PBS or the sample storage buffer as diluent. Avoid repeated freeze–thaw cycles.

FINAL EV DOSE IN ASSAY	10× WORKING SOLUTION	VOLUME ADDED TO 100 µL WELL	VOLUME ADDED TO 1 µL WELL
1 µg/mL	10 µg/mL	10 µL 10× EV + 90 µL sample/medium	100 µL 10× EV + 900 µL sample/medium
5 µg/mL	50 µg/mL	10 µL 10× EV + 90 µL sample/medium	100 µL 10× EV + 900 µL sample/medium

FINAL EV DOSE IN ASSAY	10× WORKING SOLUTION	VOLUME ADDED TO 100 µL WELL	VOLUME ADDED TO 1 mL WELL
10 µg/mL	100 µg/mL	10 µL 10× EV + 90 µL sample/medium	100 µL 10× EV + 900 µL sample/medium

Dilution formula: $C_1V_1 = C_2V_2$

Examples (stock = 1 mg/mL = 1000 µg/mL):

- To prepare 1 mL of 100 µg/mL 10× solution → 100 µL stock + 900 µL sterile PBS
- To prepare 1 mL of 50 µg/mL → 50 µL stock + 950 µL PBS
- To prepare 1 mL of 10 µg/mL → 10 µL stock + 990 µL PBS

4.5.4 Core Controls and Replicates

CONTROL / TYPE	PURPOSE	USE IN ASSAYS
Untreated cells or CP only	Baseline comparison	All assays
Vehicle control	Controls for PBS/EV buffer volume	All EV-treated assays
Mock-EV control	Controls for carryover from culture medium processed like EV extraction	EV function and cell assays
EV only	Confirms whether crude EVs are toxic by themselves	Cell assays
Challenge only	Defines CP-induced injury or inflammatory response	Cell assays
Positive assay control	Confirms assay responsiveness (validated antibacterial, oxidant/antioxidant or apoptosis control)	As applicable
Heat-inactivated EV (optional)	Tests heat-sensitive vs heat-stable EV cargo effects	Function assays

Use **at least three technical replicates** per condition and **three independent biological repeats** whenever possible.

PART



Sample Characterization Protocols

A1 BCA Protein Assay for Crude CP-Sample Concentration

Source/adaptation — Adapted from Jiang et al. (2014), Nicholas et al. (2017), Choi et al. (2020) and Mei et al. (2020), who normalized bacterial samples/MVs by protein concentration using BCA-type assays.

Purpose — To determine test sample protein concentration so all experiments can be dosed at 1, 5 and 10 $\mu\text{g/mL}$.

MATERIALS AND REAGENTS

- Crude CP-sample stock on ice
- BCA Protein Assay Kit (Thermo Scientific/Pierce or equivalent)
- BSA standards supplied with kit
- Sterile PBS or sample storage buffer
- 96-well clear flat-bottom plate
- Plate reader capable of reading 562 nm
- Pipettes and sterile low-bind tubes

ASSAY SETUP

- Run all standards and samples in duplicate or triplicate
- Prepare at least two sample dilutions (e.g. 1:5 and 1:10 in PBS) so readings fall inside the standard curve

PROCEDURE

- 1 Thaw sample aliquot on ice, mix by gentle pipetting, briefly spin down condensation
- 2 Prepare BSA standards per kit instructions — typical working range 0–2000 $\mu\text{g/mL}$
- 3 Add 25 μL of each BSA standard into assigned wells
- 4 Add 25 μL of diluted samples into assigned wells
- 5 Prepare BCA working reagent per kit instructions
- 6 Add 200 μL BCA working reagent to each well — final reaction volume 225 $\mu\text{L/well}$

- 7 Mix plate gently for **30 s** without splashing
- 8 Incubate **30 min at 37 °C** or at room temperature
- 9 Read absorbance at **562 nm**
- 10 Calculate sample concentration from the BSA standard curve and multiply by dilution factor

READOUT AND ANALYSIS

- Report sample protein concentration as $\mu\text{g}/\mu\text{L}$ and $\mu\text{g}/\text{mL}$
- Use this concentration to prepare the 10× working solutions in §4.5.3

NOTES / CRITICAL POINTS

- If the sample buffer contains detergents or high protein background, use the **same buffer as the blank**
- BCA measures total protein and does not distinguish vesicle-associated from soluble protein contamination — interpret accordingly

A2 No-Live-CP / Sterility Check of Test Sample Preparation

Source/adaptation — Based on the sterility-check concept used by Nicholas et al. (2017), who streaked purified *C. difficile* MVs onto agar/media before storage/use.

Purpose — To confirm that test sample preparations do not contain viable CP before use on chicken cells.

MATERIALS AND REAGENTS

- Crude CP-sample aliquot (or any test sample)
- Sterile PBS / vehicle control
- Mock-sample control
- CP recovery/enrichment media and plates (BHI, SFP agar)
- Anaerobic incubation
- Appropriate positive and negative controls
- **Test each sample batch before cell treatment**

PROCEDURE

- 1 In a biosafety cabinet / approved containment, test an aliquot of the sample preparation using the CP sterility method
- 2 Test the same volume of vehicle and mock-sample controls
- 3 Run the approved positive control separately to confirm the sterility test can recover viable CP
- 4 Take **10 µL** of the samples and plate on agar
- 5 Incubate **anaerobically at 37 °C for 24 hours**
- 6 Record presence/absence of growth per SOP acceptance criteria
- 7 Release sample batch for chicken-cell assays **only if no viable CP is detected**

READOUT AND ANALYSIS

- **Pass:** no viable CP detected
- **Fail:** viable CP detected — do not use that lot for chicken-cell assays unless re-purified and re-tested

NOTES / CRITICAL POINTS

- Keep a sterility record for **every** sample lot used in cell-culture experiments
- For publication, state that test samples were confirmed free of viable bacteria before cell exposure

A3 SDS-PAGE Protein Profile of Crude CP-Samples

Source/adaptation — Adapted from Jiang et al. (2014), Nicholas et al. (2017) and Mei et al. (2020), who compared MV/OMV protein profiles by SDS-PAGE and Coomassie staining.

Purpose — To compare the sample protein profile with whole-cell lysate and/or culture supernatant, and document sample lot consistency.

MATERIALS AND REAGENTS

- Crude CP-sample normalized by BCA

- CP whole-cell lysate and CP culture supernatant protein comparator (prepared under approved SOPs)
- 4× Laemmli sample buffer with reducing agent
- **12%** SDS-PAGE gel or **4–15%** gradient gel (Bio-Rad or equivalent)
- Protein ladder
- Running buffer
- Coomassie stain (e.g. PageBlue Protein Staining Solution, Thermo Scientific — used by Jiang et al. 2014 — or Bio-Rad Coomassie G-250)
- Gel imaging system

ASSAY SETUP

- Recommended loading: **10–20 µg protein/lane** for samples and whole-cell lysate (Jiang et al. loaded **20 µg** of each)
- For concentrated supernatant comparator, load an equivalent protein amount or document the volume loaded

PROCEDURE

- 1 Normalize all samples to the same protein concentration using BCA results
- 2 For each sample, prepare **20 µL total loading volume**: sample containing **10–20 µg** protein + **5 µL** 4× Laemmli buffer + PBS to **20 µL**
- 3 Denature samples according to the gel system and lab SOP
- 4 Load molecular weight marker in lane 1
- 5 Load **10–20 µg** sample protein, whole-cell lysate and culture supernatant protein in separate lanes
- 6 Run SDS-PAGE until the dye front reaches the bottom of the gel
- 7 Stain with Coomassie per reagent instructions
- 8 Destain if required and image the gel
- 9 Save image with lane labels and protein amount loaded

READOUT AND ANALYSIS

- A distinct banding pattern supports a successful sample-enriched preparation
- Compare lot-to-lot banding consistency
- Use the same samples for western blot toxin screening if needed

NOTES / CRITICAL POINTS

- SDS-PAGE alone does **not** prove vesicle purity — interpret with TEM, BCA and sterility data
- Avoid overloading test sample lanes, which obscures bands and increases background

A4 Western Blot or ELISA Screening for CP Toxins

Source/adaptation — Adapted from Jiang et al. (2014), who screened CP4 MVs for alpha toxin and NetB by western blot. Luo et al. (2020) showed CPB2 can directly injure epithelial cells, making CPB2 screening important if your strain carries *cpb2*.

Purpose — To determine whether test sample effects may be due to toxin-associated cargo or soluble toxin contamination.

MATERIALS AND REAGENTS

- Crude CP-samples, CP culture supernatant positive comparator and whole-cell lysate comparator (approved SOPs)
- SDS-PAGE and transfer system
- PVDF or nitrocellulose membrane (Jiang et al. used nitrocellulose)
- Primary antibodies: anti-alpha toxin/anti-PlcC, anti-NetB, anti-CPB2 if available — choose validated antibodies for your strain/toxin
- Secondary HRP- or AP-conjugated anti-rabbit or anti-mouse antibody
- Detection reagent: ECL for HRP, or BCIP/NBT solution (VWR) for AP — Jiang et al. used AP and BCIP/NBT
- Blocking buffer: **5%** non-fat milk in TBST, or vendor-recommended buffer
- Optional ELISA kits for alpha toxin, NetB or CPB2 if commercially available

ASSAY SETUP

- Load **10–20** µg sample protein per lane; include culture supernatant as positive comparator
- Run one membrane per toxin if antibody conditions differ

PROCEDURE

- 1 Separate samples by SDS-PAGE as in A3
- 2 Transfer proteins to membrane per your transfer system
- 3 Block with **5%** non-fat milk in TBST for **1 h at room temperature**
- 4 Incubate with toxin-specific primary antibody at vendor-recommended dilution/incubation
- 5 Wash **3× with TBST**, **5–10 min** per wash
- 6 Incubate with appropriate secondary antibody at vendor-recommended dilution
- 7 Wash **3× with TBST**
- 8 Develop signal using ECL or BCIP/NBT depending on secondary antibody enzyme
- 9 Record whether toxin bands are present in sample, supernatant and lysate fractions

READOUT AND ANALYSIS

- Toxin **not** detected in sample lane but detected in positive comparator → sample prep is less likely dominated by that toxin
- Toxin **detected** in sample lane → describe the preparation as a test sample/toxin-associated fraction and interpret cell effects cautiously

NOTES / CRITICAL POINTS

- A negative western blot does not prove complete absence of toxin if antibody sensitivity is low
- If CPB2 is detected, include CPB2-only or toxin-neutralization controls if possible

A5 Sample-Associated DNA PCR Screen (Optional)

Source/adaptation — Adapted from Jiang et al. (2014), who tested CP MV-associated DNA for 16S rRNA, *plc*, *pfoA* and *netB*, and from Mei et al. (2020), who treated OMVs with DNase I before DNA extraction and 16S PCR.

Purpose — To determine whether crude CP-samples carry CP-associated DNA and toxin gene fragments.

MATERIALS AND REAGENTS

- Crude sample
- DNase I (Thermo Fisher Scientific or equivalent; Mei et al. used **1 U**/μL)
- PBS
- Bacterial Genomic DNA Extraction Kit (Solarbio or equivalent)
- PCR primers for 16S rRNA and strain-relevant genes: *plc*, *pfoA*, *netB*, *cpb2*, *cpe*
- PCR master mix
- Thermocycler
- Agarose gel electrophoresis reagents

ASSAY SETUP

- Include: sample untreated with DNase, sample treated with DNase, CP genomic DNA positive control, and no-template control
- DNase pretreatment reduces external DNA contamination before sample disruption

PROCEDURE

- 1** Aliquot sample into two tubes: DNase-treated and untreated control
- 2** For the DNase-treated tube, add DNase I at your lab's or the manufacturer's concentration (Mei et al.: **1 U**/μL, **10 min** at **37 °C**)
- 3** Pellet or clean up samples per your lab protocol after DNase treatment
- 4** Extract DNA using a bacterial genomic DNA extraction kit
- 5** Set up PCR reactions for 16S rRNA and toxin/virulence genes relevant to your CP strain
- 6** Run agarose gel and image PCR products
- 7** Record presence/absence of amplicons

READOUT AND ANALYSIS

- 16S-positive sample DNA indicates CP-derived nucleic acid association

- *plc/pfoA/netB/cpb2/cpe* positivity supports toxin-gene DNA presence but does **not** prove toxin expression or biological activity

**NOTE — DNA PCR IS SUPPORTIVE CHARACTERIZATION ONLY.
PAIR WITH TOXIN PROTEIN TESTING AND FUNCTIONAL ASSAYS.**

A6 TEM Characterization of Crude CP-Samples

Source/adaptation — Adapted from Jiang et al. (2014), Nicholas et al. (2017), Choi et al. (2020) and Mei et al. (2020). Choi et al. diluted samples to **50 µg/mL**, placed **10 µL** on a 300-mesh copper grid and stained with **2%** uranyl acetate for **5 min**.

Purpose — To confirm sample morphology, estimate size range and assess aggregation/contamination.

MATERIALS AND REAGENTS

- Crude CP-sample
- Sterile filtered PBS
- 300- or 400-mesh carbon/Formvar-coated copper grids (EMS or equivalent)
- **2%** uranyl acetate or EM-core-approved negative stain
- Fine forceps
- Filter paper
- TEM access (instrument depends on your core facility)
- Parafilm and clean grid box

ASSAY SETUP

- Prepare sample at **50 µg/mL** in PBS as used by Choi et al.; if too dense, also prepare **25 µg/mL**
- Prepare PBS-only and mock-sample grids

PROCEDURE

- 1 Thaw sample on ice and dilute to **50 µg/mL** in sterile filtered PBS
- 2 Place a **10 µL droplet** onto the coated side of the grid

- 3 Allow adsorption for the time recommended by your EM core
- 4 Wick excess liquid gently with filter paper, without touching the center of the grid
- 5 Apply **10 μ L of 2% uranyl acetate**, stain up to **5 min** (Choi et al.) — or follow the EM core negative-staining SOP
- 6 Remove stain with filter paper and air dry completely
- 7 Image multiple random fields at suitable magnifications
- 8 Measure diameter of at least **50–100 vesicles** if image quality allows

READOUT AND ANALYSIS

- Report morphology, size range, mean diameter $\pm$ SD and representative images
- CP MVs in Jiang et al. were approximately **20–400 nm**; *C. difficile* MVs (Nicholas et al.) and *L. paracasei* EVs (Choi et al.) were spherical vesicles

NOTES / CRITICAL POINTS

- Crude preparations may show protein aggregates — TEM helps decide whether additional purification is needed
- Follow EM-core rules for stained grids containing pathogen-derived material

A7 DLS or NTA Size Distribution

Source/adaptation — Adapted from Choi et al. (2020), who diluted purified samples to **1 μ g/mL** in PBS and measured size using a Zetasizer Nano ZS.

Purpose — To quantify particle-size distribution as a supplement to TEM.

MATERIALS AND REAGENTS

- Crude CP-sample
- Sterile filtered PBS
- DLS instrument (Malvern Zetasizer Nano ZS) or NTA instrument (NanoSight / ZetaView)
- Low protein-binding tubes
- Disposable cuvettes or NTA sample chamber materials

ASSAY SETUP

- Prepare dilution at **1 µg/mL** in filtered PBS for DLS (Choi et al.); prepare additional dilutions if the count rate is too high

PROCEDURE

- 1 Dilute sample stock to **1 µg/mL** in sterile filtered PBS
- 2 Centrifuge briefly at low speed **only if needed** to remove large dust particles — avoid pelleting vesicles
- 3 Load sample into DLS cuvette or NTA chamber per instrument SOP
- 4 Run at least **three technical measurements** per sample lot
- 5 Record hydrodynamic diameter and polydispersity index (DLS) or particle concentration (NTA)

READOUT AND ANALYSIS

- Report modal/mean size, polydispersity and particle concentration if available
- Use TEM as the main morphology confirmation, because crude bacterial vesicle samples may be polydisperse

NOTES / CRITICAL POINTS

- DLS is sensitive to large aggregates and may **overestimate** size in test samples
- NTA provides particle concentration but requires appropriate dilution and instrument access

PART

B

Antibacterial and Anti- Virulence Protocols Against CP

CRITICAL

Important: The assays below are written as assay layouts and endpoint procedures. Prepare and handle live CP **only** according to the laboratory SOP. Do not use these notes as a stand-alone protocol for propagating CP.

Standard Treatment Layout

GROUP	PER 1 ML ASSAY TUBE/WELL FINAL MIXTURE	FINAL EV DOSE
B1 CP only	900 µL standardized CP suspension + 100 µL medium	0 µg/mL
B2 vehicle	900 µL standardized CP suspension + 100 µL EV PBS	0 µg/mL
B3 mock-EV	900 µL standardized CP suspension + 100 µL 10× mock-EV	matched
B4 EV-low	900 µL standardized CP suspension + 100 µL 10 µg/mL EV	1 µg/mL
B5 EV-mid	900 µL standardized CP suspension + 100 µL 50 µg/mL EV	5 µg/mL
B6 EV-high	900 µL standardized CP suspension + 100 µL 100 µg/mL EV	10 µg/mL
B7 positive control	900 µL standardized CP suspension + 100 µL validated positive-control reagent	as validated

B1 CP Growth-Readout Assay with Sample Treatment

Source/adaptation — Conceptually based on bacterial vesicle functional assays in Jiang et al. (2014), Nicholas et al. (2017) and Mei et al. (2020), adapted here to test whether CP-derived test samples inhibit or promote CP growth. Live CP preparation must follow the lab SOP.

Purpose — To determine whether 1, 5 or 10 µg/mL test sample alters CP growth kinetics.

MATERIALS AND REAGENTS

- Standardized CP suspension prepared by the laboratory CP SOP

- Crude CP-sample 10× working solutions: 10, 50 and **100 µg/mL**
- PBS / sample vehicle
- Mock-sample preparation
- Validated positive antibacterial control
- Anaerobe-compatible microplate, sealed plate system or approved tubes
- Plate reader for OD₆₀₀ (microplate format)
- Sterile pipette tips and reservoirs

ASSAY SETUP

- Recommended final volume: **1 mL/tube** or **200 µL/well**
- For **200 µL/well**: **180 µL** standardized CP suspension + **20 µL** 10× sample/control
- For **1 mL/tube**: **900 µL** CP suspension + **100 µL** 10× sample/control
- Use at least **triplicate** wells/tubes per group

PROCEDURE

- 1 Prepare the standardized CP suspension using your approved lab SOP
- 2 Label tubes/wells: CP only, vehicle, mock-sample, sample 1 / 5 / **10 µg/mL**, positive control
- 3 Add CP suspension first — **180 µL** per **200 µL** well, or **900 µL** per **1 mL** tube
- 4 Add 10× sample or control — **20 µL** per **200 µL** well, or **100 µL** per **1 mL** tube
- 5 Mix gently by pipetting or gentle plate shaking, without creating aerosols
- 6 Incubate using the laboratory anaerobic CP assay conditions
- 7 Measure OD₆₀₀ at baseline and selected time points (e.g. 0, 6, 12, **24 h**)
- 8 At endpoint, collect aliquots for viable-count confirmation using SOP B2

READOUT AND ANALYSIS

- Plot OD₆₀₀ vs time
- **Percent growth inhibition = [(OD control – OD treated) / OD control] × 100**
- Confirm any OD-based effect by CFU, because sample particles can influence turbidity

NOTES / CRITICAL POINTS

- If samples increase OD **without** increasing CFU, the apparent effect may be particle aggregation rather than bacterial growth
- If samples increase **both** OD and CFU, samples may **promote** CP growth rather than inhibit it

B2 Viable-Count Confirmation After Sample Treatment

Source/adaptation — Added as a required confirmation for OD-based antibacterial results; viability confirmation matters especially because test samples may affect turbidity.

Purpose — To determine whether sample treatment reduces viable CP burden.

MATERIALS AND REAGENTS

- Endpoint samples from B1
- Sterile diluent approved for CP enumeration
- Laboratory CP plating/enumeration materials
- Sterile tubes · Pipettes
- Approved anaerobic incubation system
- Colony counter or manual counting sheet

ASSAY SETUP

- Use the same treatment groups as B1
- Use the dilution series, plating volume and incubation conditions already validated in your lab for CP enumeration

PROCEDURE

- 1 At the selected endpoint, gently mix each sample
- 2 Remove a defined aliquot (e.g. **100 µL**) for viable-count processing
- 3 Perform serial dilution using the laboratory-approved CP enumeration SOP
- 4 Plate the validated volume from appropriate dilutions
- 5 Incubate under approved CP conditions

- 6 Count colonies from countable dilution plates
- 7 Calculate CFU/mL and $\log_{10}$ CFU/mL for each sample

READOUT AND ANALYSIS

- **Log reduction = $\log_{10}(\text{CFU/mL control}) - \log_{10}(\text{CFU/mL treated})$**
- A decrease in $\log_{10}$ CFU/mL supports antibacterial activity
- No CFU reduction but decreased toxin gene expression suggests **anti-virulence** rather than killing

NOTES / CRITICAL POINTS

- Do not report OD₆₀₀ alone as antibacterial activity
- Include the mock-sample control, because medium-derived carryover can influence bacterial growth

B3 Agar Well Diffusion Assay

Added as a screening assay for antimicrobial activity of test samples against CP. Standardized CP suspension (**1.0×10^6 concentration preferred**), recovery conditions and containment procedures must follow the laboratory SOP.

Purpose — To determine whether test samples produce a measurable zone of growth inhibition on an inoculated agar surface.

MATERIALS AND REAGENTS

- Standardized CP suspension per laboratory CP SOP (check McFarland SOP)
- Laboratory-validated agar plates suitable for CP testing
- Sterile swabs or validated plate-inoculation device
- Sterile **6 mm** well borer or equivalent
- Test-sample concentration series, including 1, 5 and **10 $\mu\text{g/mL}$** where suitable
- Vehicle, mock-sample and validated positive antibacterial controls
- Sterile pipettes and tips

- Caliper or ruler for zone measurement
- Anaerobic incubation system

ASSAY SETUP

- Use the same standardized inoculum for all plates
- Use equal well diameter and equal sample volume for every treatment
- Run **at least three independent wells** per treatment
- Record both the sample concentration and total mass loaded per well

PROCEDURE

- 1 Inoculate the complete agar surface uniformly with the standardized CP suspension (1.0×10^6)
- 2 Allow excess surface moisture to absorb before making wells
- 3 Create wells of equal diameter using a cork borer (4–9 mm) and remove agar plugs carefully
- 4 Add an equal validated volume — commonly 50–100 μL — of each test sample or control to the assigned well (depends on borer diameter)
- 5 Keep wells sufficiently separated to prevent overlapping inhibition zones
- 6 Use the same pre-diffusion period for all plates
- 7 Incubate using laboratory CP assay conditions
- 8 Measure the **complete inhibition-zone diameter in mm, including the well diameter**
- 9 Record irregular, partial or indistinct zones; photograph plates with a scale where possible

READOUT AND ANALYSIS

- Report mean inhibition-zone diameter $\pm$ SD for each treatment
- A larger clear zone indicates greater inhibition under the specific diffusion conditions used
- Compare treatments only when inoculum, agar depth, well size, sample volume and incubation conditions are identical

NOTES / CRITICAL POINTS

- **A lack of a zone does not prove lack of antimicrobial activity** — large proteins, particles and hydrophobic materials diffuse poorly through agar

- Agar well diffusion is a **screening** method — confirm with MIC, MBC and/or CFU assays
- Do not assign clinical susceptibility categories to nonstandard test samples

B4 Disk Diffusion Assay

Source/adaptation — Added as a standardized plate-based screening approach based on disk-diffusion principles. For experimental test samples, report **raw zone diameters** rather than clinical susceptibility categories.

Purpose — To screen whether a defined quantity of test sample inhibits CP growth around a sample-loaded disk.

MATERIALS AND REAGENTS

- Standardized CP suspension per laboratory CP SOP
- Laboratory-validated agar plates
- Sterile **6 mm** blank paper disks or validated commercial disks
- Sterile forceps or disk dispenser
- Test-sample concentration series
- Vehicle-loaded blank disks
- Validated positive antibacterial control disk
- Caliper or ruler
- Anaerobic incubation system

ASSAY SETUP

- Use identical blank disks and loading volume for all experimental samples
- Report total mass of test sample loaded per disk
- Allow loaded disks to absorb the sample under sterile conditions before placement
- Use at least **three disks or independent plates** per treatment

PROCEDURE

- 1** Inoculate the agar surface uniformly with the standardized CP suspension

- 2 Allow excess surface moisture to absorb
- 3 Load blank disks with the validated volume of test sample, vehicle or control
- 4 Place disks on the inoculated agar using sterile forceps; press gently for complete contact
- 5 **Do not move disks** after they contact the agar
- 6 Space disks sufficiently to prevent overlapping zones
- 7 Incubate using laboratory CP assay conditions
- 8 Measure inhibition-zone diameter in mm
- 9 Record colonies within zones, double zones or indistinct edges separately

READOUT AND ANALYSIS

- Report mean zone diameter $\pm$ SD and the amount of sample loaded per disk
- For experimental sample-loaded disks, report **raw zone diameters** without S/I/R classification
- Confirm important findings using MIC, MBC and CFU testing

NOTES / CRITICAL POINTS

- Disk diffusion depends strongly on sample diffusion and may underestimate poorly soluble or particulate materials
- Use the same agar depth, inoculum, disk size, loading volume and incubation conditions across comparisons
- Commercial antibiotic breakpoints must not be applied to nonstandard test-sample disks

B5 Minimum Inhibitory Concentration (MIC) Assay

Added as a broth-microdilution endpoint to quantify the lowest test-sample concentration preventing detectable CP growth under laboratory assay conditions.

Purpose — To determine the lowest concentration of test sample that inhibits detectable CP growth.

MATERIALS AND REAGENTS

- Standardized CP suspension per laboratory CP SOP
- Laboratory-validated broth medium

- Sterile clear 96-well microplate or validated assay tubes
- Test-sample stock and serial dilution series
- Vehicle, mock-sample and positive antibacterial controls
- Sample-only blank wells without CP
- Plate reader for OD₆₀₀ when optical measurement is used
- Sterile pipettes, tips and reservoirs

ASSAY SETUP

- Use a **two-fold serial dilution** bracketing the inhibitory endpoint whenever possible
- The 1, 5 and **10 µg/mL** study concentrations may be included, but a broader series may be needed to establish a formal MIC
- For a **200 µL** final well volume, one practical layout is **100 µL of 2× test sample + 100 µL standardized CP suspension**
- Run at least **triplicate** wells per concentration

PROCEDURE

- 1 Prepare the test-sample concentration series in the selected broth
- 2 Dispense sample dilutions into assigned wells
- 3 Add standardized CP suspension to treatment, growth-control and vehicle-control wells
- 4 Add sterile broth without CP to sample-only and sterility-control wells
- 5 Keep final volume and vehicle concentration equal in all wells
- 6 Mix gently without producing bubbles or aerosols
- 7 Record baseline OD₆₀₀ if an optical endpoint is used
- 8 Incubate using laboratory CP assay conditions
- 9 Assess growth visually and/or using the validated OD₆₀₀ endpoint
- 10 Subtract the matching sample-only blank from colored, turbid or particulate sample readings
- 11 Record the lowest concentration meeting the predefined no-growth criterion as the **MIC**
- 12 Confirm the MIC result by viable-count or MBC testing

READOUT AND ANALYSIS

- Report MIC in $\mu\text{g/mL}$ or the appropriate sample concentration unit
- If tested concentrations do not bracket complete inhibition, report as **> highest tested concentration** or as the lowest tested inhibitory concentration — do not assign an unsupported MIC
- Present replicate MIC values and the modal value across independent experiments

NOTES / CRITICAL POINTS

- MIC indicates **inhibition**, not killing
- Test-sample color, turbidity or particles may interfere with OD — use sample-only blanks and viable-count confirmation
- Growth and sterility controls must perform as expected before the plate is interpreted

B6 Minimum Bactericidal Concentration (MBC) Assay

Added as a viable-recovery assay following MIC testing, to determine whether inhibitory concentrations are bactericidal.

Purpose — To identify the lowest test-sample concentration producing the laboratory-defined bactericidal endpoint.

MATERIALS AND REAGENTS

- Completed MIC plate or tubes
- Laboratory-validated recovery agar
- Sterile diluent
- Sterile pipettes and tips
- Anaerobic incubation system
- Colony-counting materials

ASSAY SETUP

- Use the MIC well and all higher concentrations showing no detectable growth

- Include the untreated or vehicle growth control and sterility control
- Use dilution, neutralization or washing where required to minimize test-sample carryover onto recovery agar

PROCEDURE

- 1 Identify the MIC well and higher-concentration wells without detectable growth
- 2 Mix each selected well gently
- 3 Remove a defined aliquot from each selected concentration
- 4 Dilute or neutralize the aliquot per the validated recovery method
- 5 Plate the validated volume onto recovery agar
- 6 Process a growth-control sample in parallel
- 7 Incubate under laboratory CP recovery conditions
- 8 Count recovered colonies
- 9 Determine the lowest concentration meeting the predefined bactericidal criterion
- 10 **Report MIC and MBC together**

READOUT AND ANALYSIS

- A commonly used bactericidal endpoint is a reduction of $\geq 99.9\%$ relative to the starting or growth-control inoculum — but the exact criterion used must be stated
- Report MBC in the same units as MIC
- MBC/MIC ratio may be reported as a descriptive comparison

NOTES / CRITICAL POINTS

- Residual test sample transferred to agar may suppress colony recovery and create a **false bactericidal result** — validate dilution or neutralization
- If colonies are recovered at every concentration, report that MBC was above the highest concentration tested
- Use CFU data rather than optical density to determine bactericidal activity

B7 Bacterial Live/Dead Staining After Sample Treatment

Source/adaptation — Functional endpoint added to test membrane integrity; complements LDH/cytotoxicity concepts used by Jiang et al. (2014) and Luo et al. (2020).

Purpose — To visualize whether sample treatment damages CP bacterial membranes.

MATERIALS AND REAGENTS

- Sample-treated CP samples from B1
- LIVE/DEAD BacLight Bacterial Viability Kit (Thermo Fisher/Invitrogen or equivalent)
- Sterile PBS
- Black 96-well plate or microscope slides
- Fluorescence microscope or flow cytometer
- Validated positive dead-cell control

ASSAY SETUP

- Use final sample doses **0, 1, 5 and 10** µg/mL
- Protect dyes from light. Use the bacterial staining kit exactly as directed by the manufacturer

PROCEDURE

- 1 At endpoint, collect a small aliquot of each treatment sample under containment
- 2 Wash or dilute samples in sterile PBS per kit instructions
- 3 Prepare live/dead dye mixture per kit instructions
- 4 Mix sample with dye mixture in the manufacturer-recommended volume ratio
- 5 Incubate protected from light for the recommended time
- 6 Transfer stained sample to slide or black 96-well plate
- 7 Image using green/red fluorescence channels, or analyze by flow cytometry
- 8 Quantify live/dead ratio using ImageJ, plate-reader fluorescence or cytometry gating

READOUT AND ANALYSIS

- Increased red/dead signal in sample-treated CP suggests **membrane damage**
- No change in live/dead staining despite lower toxin gene expression suggests a **non-bactericidal anti-virulence effect**

NOTES / CRITICAL POINTS

- Some anaerobes and dense samples may stain unevenly — optimize using positive live and dead controls
- Check sample autofluorescence/background using sample-only dye controls

B8 CP Biofilm Inhibition / Disruption Assay (Crystal Violet Readout)

Source/adaptation — Added to evaluate sample effects on CP attachment/biomass. Not directly from one uploaded paper, but complements the vesicle functional analysis framework used by Jiang et al. (2014), Nicholas et al. (2017) and Mei et al. (2020).

Purpose — To test whether crude CP-samples prevent biofilm formation or disrupt established CP biofilm.

MATERIALS AND REAGENTS

- Standardized CP suspension per approved SOP
- Crude sample 10× working solutions
- PBS/vehicle and mock-sample
- Sterile flat-bottom 96-well plate compatible with biofilm assay
- Crystal violet solution (commonly 0.1% w/v or lab-validated concentration)
- PBS
- Solubilization solution: 30% acetic acid or lab-validated alternative
- Plate reader for 570–590 nm
- Approved disinfectant/waste system for CP biofilm material

ASSAY SETUP

- **Biofilm inhibition format:** add sample at time zero
- **Biofilm disruption format:** establish biofilm per approved SOP, gently wash, then add sample treatments

- Suggested final volume **200 µL/well**: **180 µL** CP suspension/medium + **20 µL** 10× sample/control

PROCEDURE

- 1 Prepare treatment plate with at least **triplicate** wells per group
- 2 For inhibition assay, add **180 µL** standardized CP suspension to each well
- 3 Add **20 µL** 10× sample/control to reach 0, 1, 5 or **10 µg/mL** final concentration
- 4 Incubate using the approved CP biofilm SOP
- 5 At endpoint, remove planktonic suspension carefully without disturbing attached biofilm
- 6 Wash wells gently with PBS using the lab-approved containment method
- 7 Fix biofilm if required by your SOP before staining
- 8 Add **200 µL crystal violet** per well and incubate anaerobically
- 9 Remove stain and wash gently until background is low
- 10 Add **200 µL 30% acetic acid** (or validated solubilization solution) per well
- 11 Transfer **100–150 µL** solubilized stain to a clean plate if needed
- 12 Read absorbance at **570–590 nm**

READOUT AND ANALYSIS

- **Biofilm biomass (%) = (OD treated / OD CP control) × 100**
- Reduced crystal violet staining suggests reduced attached biomass
- Pair with biofilm CFU or SEM — crystal violet does **not** measure viability

NOTES / CRITICAL POINTS

- Crystal violet is sensitive to washing technique — use consistent gentle washing
- If the strain forms weak biofilms, this assay may require optimization by your lab

B9 CP Virulence / Toxin Gene-Expression qRT-PCR After Sample Treatment

Source/adaptation — Gene-target selection based on Jiang et al. (2014), who examined CP MV-associated 16S rRNA, *plc*, *pfoA* and *netB*, and Luo et al. (2020), who measured inflammatory/apoptosis/tight-junction gene expression by qRT-PCR using the $2^{-\Delta\Delta C_t}$ method.

Purpose — To determine whether test samples suppress or increase CP virulence gene expression independent of bacterial killing.

MATERIALS AND REAGENTS

- Sample-treated CP samples from B1
- Bacterial RNA extraction kit suitable for **Gram-positive** bacteria
- DNase treatment reagent
- cDNA synthesis kit (e.g. PrimeScript RT reagent Kit with gDNA Eraser, Takara — used by Luo et al. for cell RNA)
- SYBR qPCR master mix (e.g. SYBR Premix Ex Taq, Takara, or Applied Biosystems SYBR Green qPCR Mastermix)
- Primers for CP genes: 16S rRNA / *rpoA* / *gyrA* housekeeping; *plc*, *netB*, *cpb2*, *pfoA*, *cpe* if strain-relevant
- Real-time PCR system

ASSAY SETUP

- Harvest RNA at a time point **before major death occurs**, or normalize carefully to CFU
- Use **three biological replicates** and **technical qPCR triplicates**

PROCEDURE

- 1 Collect treated bacterial samples using the laboratory CP handling SOP
- 2 Pellet/lyse samples using the bacterial RNA kit instructions validated for Gram-positive bacteria
- 3 Perform DNase treatment to remove genomic DNA
- 4 Quantify RNA and assess purity
- 5 Reverse transcribe **equal RNA input per sample** (e.g. 500 ng – 1 µg RNA depending on kit and yield)

- 6 Prepare **20 μ L qPCR reaction** (Luo et al. style): **10 μ L** 2 $\times$ SYBR mix + **0.8 μ L** forward/reverse primer mix (or **0.4 μ L** each primer) + **2 μ L** cDNA + nuclease-free water to **20 μ L**. Adjust primer volume to your stock concentration
- 7 Run qPCR using validated primer annealing conditions
- 8 Analyze relative expression using **$2^{-\Delta\Delta Ct}$**

READOUT AND ANALYSIS

- Report fold change vs CP-only or vehicle control
- Reduced *plc/netB/cpb2/pfoA/cpe* expression suggests an **anti-virulence effect**
- Increased expression suggests samples may stimulate CP stress response or virulence signaling

NOTES / CRITICAL POINTS

- Use RNA extraction methods that effectively lyse Gram-positive CP
- Interpret gene expression **with CFU data** — lower biomass may reduce apparent transcript abundance

PART

C

Chicken-Cell Protection Protocols

Recommended first model: chicken intestinal epithelial cells with sterile CP challenge material for initial protection assays. HD11 chicken macrophages are useful for inflammatory gene studies — Mei et al. (2020) used HD11 cells and measured IL-1 β , IL-6, IL-2, iNOS and IL-10 after OMV exposure.

Standard Treatment Design (96-well assays)

GROUP	TREATMENT	PURPOSE
C1 untreated	90 μ L medium + 10 μ L PBS; no challenge	Baseline
C2 vehicle	90 μ L medium + 10 μ L EV buffer; no challenge	Vehicle control
C3 mock-EV	90 μ L medium + 10 μ L 10 $\times$ mock-EV; no challenge	Mock-process control
C4 EV 1 only	90 μ L medium + 10 μ L 10 μ g/mL EV	EV-only cytotoxicity
C5 EV 5 only	90 μ L medium + 10 μ L 50 μ g/mL EV	EV-only cytotoxicity
C6 EV 10 only	90 μ L medium + 10 μ L 100 μ g/mL EV	EV-only cytotoxicity
C7 challenge only	Cells + CP challenge material, no EV	Injury control
C8 EV 1 + challenge	EV pretreatment at 1 μ g/mL then challenge	Protection
C9 EV 5 + challenge	EV pretreatment at 5 μ g/mL then challenge	Protection
C10 EV 10 + challenge	EV pretreatment at 10 μ g/mL then challenge	Protection

C1 Sample-Only Cytotoxicity Screen in Chicken Cells

Source/adaptation — Adapted from Choi et al. (2020), who tested *L. paracasei* EV cell viability at doses including 1 and 10 μ g/mL, and from Luo et al. (2020), who used MTT to evaluate epithelial-cell viability.

Purpose — To identify whether crude CP-samples are toxic to chicken cells before challenge experiments.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells or HD11 chicken macrophage cells
- Complete cell culture medium appropriate for the cell type
- Crude CP-sample 10× working solutions
- PBS/vehicle and mock-sample
- 96-well clear plate
- Cell viability reagent: CCK-8/WST-8 kit (DoGen Bio Cell Viability Assay Kit as used by Choi et al.), MTT reagent (Solarbio, used by Luo et al.), or Promega CellTiter 96/CellTiter-Glo
- Plate reader — **450 nm** for CCK-8/WST-8, **570 nm** for MTT

ASSAY SETUP

- Suggested seeding density: one giving **70–90% confluence at treatment endpoint**; for many epithelial cells, start with **1×10^4 cells/well in 100 μL** (Luo et al. used 1×10^4 IPEC-J2 cells/well for MTT)
- Final treatment volume: **100 μL /well**

PROCEDURE

- 1 Seed **100 μL** cell suspension per well in a 96-well plate; allow cells to attach overnight
- 2 Prepare 10× sample working solutions at 10, 50 and **100 $\mu\text{g/mL}$**
- 3 Remove old medium carefully if required and add **90 μL fresh medium** per well
- 4 Add **10 μL** PBS/vehicle, mock-sample or 10× sample solution per well → 0, 1, 5 or **10 $\mu\text{g/mL}$** final
- 5 Incubate for **12 h and/or 24 h** (Choi et al. used **12 h** EV treatment for viability; Luo et al. used **24 h** toxin treatment)
- 6 Add viability reagent per kit instructions. For CCK-8/WST-8: add **10 μL** reagent to **100 μL** culture medium per well and incubate until color develops (Choi et al. incubated **3 h**, read at **450 nm**)
- 7 Read absorbance at the kit-recommended wavelength
- 8 Calculate viability relative to untreated cells

READOUT AND ANALYSIS

- **Cell viability (%) = (Abs treated / Abs untreated control) × 100**
- Proceed with doses showing acceptable sample-only viability, ideally $\geq 90\%$
- Interpret doses $< 80\%$ as cytotoxic

NOTES / CRITICAL POINTS

- Include blank wells containing **medium + reagent without cells**
- Use phenol red/serum conditions compatible with your selected viability kit

C2 Cell Viability Protection Assay After CP Challenge

Source/adaptation — Adapted from Luo et al. (2020) for MTT viability, and Choi et al. (2020) for EV pretreatment before inflammatory challenge.

Purpose — To test whether crude CP-sample pretreatment improves chicken-cell viability after CP challenge.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells
- Crude sample 10× working solutions
- Sterile CP challenge material prepared by laboratory SOP — preferably filtered CP supernatant/toxin fraction for first-round studies
- PBS/vehicle and mock-sample
- 96-well clear plate
- MTT, CCK-8 or WST-8 viability kit
- DMSO if using MTT (Luo et al. dissolved formazan in **100 μ L** DMSO per well)
- Plate reader

ASSAY SETUP

- Final sample pretreatment volume: **100 μ L/well**
- Use sample pretreatment for **12 h** (Choi et al. style design), then challenge for **12–24 h**

- Do not include antibiotics during live CP challenge unless your approved SOP requires otherwise; for sterile CP supernatant challenge, maintain lab-validated cell culture conditions

PROCEDURE

- 1 Seed cells in 96-well plate (e.g. 1×10^4 cells/well in 100 μ L) and culture overnight
- 2 Prepare treatment medium: 90 μ L **fresh medium** + 10 μ L **10 $\times$ sample/control** per well $\rightarrow$ 1, 5 or 10 μ g/mL
- 3 Pretreat cells with sample/control for 12 h
- 4 Add CP challenge material using the determined volume and concentration (**MOI of 1**). Keep final volume consistent across wells
- 5 Include challenge-only wells and no-challenge sample-only wells
- 6 Incubate 12–24 h depending on injury kinetics
- 7 **CCK-8/WST-8**: add 10 μ L reagent per 100 μ L culture volume, incubate per kit, read 450 nm
- 8 **MTT (Luo et al.)**: add 10 μ L MTT per well, incubate 4 h, remove medium carefully, add 100 μ L **DMSO** per well, incubate 10 min **protected from light**, read 570 nm
- 9 Calculate percent viability relative to untreated control

READOUT AND ANALYSIS

- Protection is supported if CP challenge lowers viability and sample pretreatment **restores** viability
- Plot viability (%) for untreated, challenge-only, sample-only and sample + challenge groups

NOTES / CRITICAL POINTS

- Run an **injury pilot first** to choose a CP challenge condition that reduces viability without killing nearly all cells — a **30–50% reduction** is useful for detecting protection
- If test samples alone are toxic, protection results may be difficult to interpret

C3 LDH Release Assay for Cytotoxicity / Membrane Damage

Source/adaptation — Adapted from Luo et al. (2020), who seeded 1×10^5 cells/well in 24-well plates, treated for 24 h, collected supernatant and read LDH at 450 nm using a Jiancheng kit. Jiang et al. (2014) also used a Biovision LDH kit for MV cytotoxicity.

Purpose — To determine whether sample pretreatment reduces CP-induced cell membrane damage.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells
- 24-well plate
- Sample 10× working solutions
- CP challenge material per approved SOP
- LDH Cytotoxicity Assay Kit: Jiancheng (Nanjing, China), Biovision, or Promega CytoTox 96
- Plate reader capable of 450 nm or kit-specified wavelength
- Sterile PBS and complete medium
- Positive lysis control supplied with kit

ASSAY SETUP

- Suggested seeding density: 1×10^5 cells/well in 500 μL – 1 mL medium in 24-well plates (adapted from Luo et al.)
- Treatment volume: 1 mL/well is convenient — 900 μL medium + 100 μL 10× sample/control

PROCEDURE

- 1 Seed 1×10^5 cells/well in a 24-well plate; incubate overnight
- 2 Prepare 10× sample/control treatments
- 3 Replace medium with 900 μL fresh medium per well
- 4 Add 100 μL 10× sample/control → 0, 1, 5 or 10 $\mu\text{g/mL}$
- 5 Pretreat cells for 12 h
- 6 Add CP challenge material per the pilot injury condition; keep final volume equal in all wells
- 7 Incubate 12–24 h
- 8 Collect culture supernatant into labeled tubes; clarify gently if kit recommends

- 9 Transfer the kit-specified volume of supernatant into a 96-well assay plate — commonly **50–100 µL/well**
- 10 Add LDH reaction mixture per kit instructions
- 11 Incubate protected from light per kit instructions
- 12 Read absorbance at **450 nm** or kit-specified wavelength

READOUT AND ANALYSIS

CALCULATION

LDH release (%) = [(experimental LDH – spontaneous LDH) / (maximum LDH – spontaneous LDH)] × 100 *(when a maximum lysis control is included)*

Protection is supported if challenge increases LDH and sample pretreatment reduces LDH release.

NOTES / CRITICAL POINTS

- Avoid disturbing cells before collecting supernatant
- If using live CP challenge, ensure supernatants are handled/inactivated per biosafety SOP before plate-reader work

C4 DCFDA ROS Assay for Oxidative Stress

Source/adaptation — Adapted from Luo et al. (2020), who used 10 µM DCFH-DA, **1 mL/well** in 6-well plates, incubated **20 min** at **37 °C** and read fluorescence at Ex/Em 488/**525 nm**. User-selected kit: Abcam DCFDA/H2DCFDA Cellular ROS Assay Kit.

Purpose — To test whether sample pretreatment reduces CP-induced intracellular ROS in chicken cells.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells
- 6-well plate for microscopy/flow, or black 96-well plate for plate-reader quantification
- Abcam DCFDA/H2DCFDA Cellular ROS Assay Kit or Solarbio DCFH-DA

- Serum-free medium or kit assay buffer
- Sample 10× working solutions
- CP challenge material per approved SOP
- Positive ROS control supplied/recommended by kit (e.g. tert-butyl hydroperoxide if lab-approved)
- Fluorescence plate reader or microscope with FITC/GFP channel — **Ex/Em $\approx$ 488/525 nm**

ASSAY SETUP

- **6-well format:** seed 1×10^5 cells/well in 2 mL medium (as in Luo et al. for ROS experiments)
- **96-well quantification:** seed optimized density, typically $1-2 \times 10^4$ cells/well in 100 μ L

PROCEDURE

- 1 Seed cells and allow attachment overnight
- 2 Pretreat with samples at 1, 5 and 10 μ g/mL for 12 h using 10× additions
- 3 Challenge cells with CP challenge material for the selected injury time
- 4 Remove treatment medium carefully
- 5 Wash cells once with PBS or assay buffer
- 6 Prepare DCFDA working solution per Abcam kit. Following Luo et al., use final 10 μ M DCFH-DA
- 7 6-well plate: add 1 mL of 10 μ M DCFDA working solution per well. 96-well plate: add 100 μ L per well
- 8 Incubate 20 min at 37 °C **protected from light** (Luo et al.); use kit instructions if different
- 9 Wash cells 1–2× with 100 μ L serum-free medium or assay buffer to remove extracellular dye
- 10 Read fluorescence at **Ex/Em 488/525 nm**, or image with a fluorescence microscope
- 11 Normalize fluorescence to viable cell number if CP challenge strongly changes cell density

READOUT AND ANALYSIS

- **ROS fold change = fluorescence treated / fluorescence untreated control**
- Protection is supported if CP challenge increases ROS and sample pretreatment decreases ROS

NOTES / CRITICAL POINTS

- DCFDA is **light-sensitive** and can auto-oxidize — prepare fresh working solution
- Include sample-only + DCFDA wells to check sample-related fluorescence/background

C5 JC-1 Mitochondrial Membrane Potential Assay

Source/adaptation — Adapted from Luo et al. (2020), who added **1 mL** fresh medium plus **1 mL** JC-1 working solution per well, incubated **20 min** at **37 °C** and imaged at 200×

Purpose — To determine whether sample pretreatment preserves mitochondrial membrane potential after CP challenge.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells
- 6-well or 24-well plate
- JC-1 Mitochondrial Membrane Potential Assay Kit (Beyotime — used by Luo et al. — or Abcam/Thermo equivalent)
- **JC-1** staining buffer
- Sample working solutions
- CP challenge material per approved SOP
- Fluorescence microscope or fluorescence plate reader

ASSAY SETUP

- Use 6-well format for microscopy, or 96-well format for quantitative plate-reader assay depending on kit
- For 6-well format, maintain enough cells for imaging without over-confluence

PROCEDURE

- 1 Seed cells and allow attachment overnight
- 2 Pretreat cells with samples at 1, 5 and **10 µg/mL** for **12 h**
- 3 Apply CP challenge material for the selected period

- 4 Remove old medium and wash cells once with PBS
- 5 6-well format (Luo et al.): add **1 mL fresh culture medium + 1 mL JC-1 working solution** per well
- 6 Incubate **20 min at 37 °C protected from light**
- 7 Remove staining solution and wash **twice with pre-cooled 1× JC-1 staining buffer**
- 8 Image red and green fluorescence channels, or quantify with plate reader per kit
- 9 Calculate **red/green fluorescence ratio**

READOUT AND ANALYSIS

- Healthy mitochondria → higher **red** JC-1 aggregate signal
- Damaged mitochondria → higher **green** monomer signal
- Protection is supported if sample pretreatment **restores the red/green ratio** compared with CP challenge alone

NOTES / CRITICAL POINTS

- Use the **same exposure settings** across all microscopy groups
- A strong decrease in cell number may confound fluorescence — pair with viability assay

C6 Annexin V-FITC / PI Apoptosis Assay

Source/adaptation — Adapted from Luo et al. (2020), who used an Annexin V-FITC/PI kit from Beyotime; cells were resuspended in **195 µL** binding buffer and stained with **5 µL** Annexin V-FITC plus **10 µL** PI for **20 min** in the dark.

Purpose — To assess whether sample pretreatment reduces CP-induced apoptosis / late cell death.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells
- 6-well plate
- Annexin V-FITC/PI Apoptosis Detection Kit (Beyotime, BD Biosciences, Abcam or Thermo Fisher)
- Cold PBS

- Binding buffer supplied with kit
- Sample 10× working solutions
- CP challenge material per approved SOP
- Flow cytometer

ASSAY SETUP

- Use enough cells per condition for flow cytometry — 6-well plates recommended
- **Collect both floating and adherent cells** to avoid underestimating dead/apoptotic cells

PROCEDURE

- 1 Seed cells in 6-well plates; allow attachment overnight
- 2 Pretreat with samples at 1, 5 and **10 µg/mL** for **12 h**
- 3 Challenge cells with CP challenge material for the selected injury period
- 4 Collect culture supernatant containing floating cells into labeled tubes
- 5 Detach adherent cells gently using the cell-type-appropriate method; combine with floating cells
- 6 Wash cells **twice with cold PBS**
- 7 Resuspend cell pellet in **195 µL binding buffer** per sample (Luo et al.)
- 8 Add **5 µL Annexin V-FITC** and **10 µL PI** per sample
- 9 Incubate **20 min at room temperature in the dark**
- 10 Analyze **promptly** by flow cytometry
- 11 Record live, early apoptotic, late apoptotic and necrotic populations

READOUT AND ANALYSIS

- Protection is supported if sample pretreatment reduces Annexin V-positive and/or PI-positive fractions compared with CP challenge alone
- Report representative dot plots and quantified percentages

NOTES / CRITICAL POINTS

- Set compensation using **single-stained controls**

- If live CP was used, confirm samples are handled per cytometer biosafety requirements

C7 qRT-PCR for Inflammatory, Oxidative-Stress, Barrier and Apoptosis Genes

Source/adaptation — Adapted from Luo et al. (2020), Choi et al. (2020) and Mei et al. (2020). Luo et al. used TRIzol, PrimeScript RT reagent Kit with gDNA Eraser, SYBR Premix Ex Taq and **20 µL** qPCR reactions. Mei et al. used HD**11 cells** and measured chicken IL-1 β , IL-6, IL-2, iNOS and IL-10.

Purpose — To define whether sample pretreatment reduces inflammation/oxidative stress and preserves barrier and survival pathways.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells or HD**11 cells**
- 6-well plate
- TRIzol reagent (Invitrogen or TransGen Biotech — Luo used TransGen; Choi/Mei used Invitrogen)
- cDNA kit: PrimeScript RT reagent Kit with gDNA Eraser (Takara) or RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific)
- SYBR Premix Ex Taq (Takara) or Applied Biosystems SYBR Green qPCR Mastermix
- Primers for chicken genes of interest
- Nuclease-free water
- Real-time PCR instrument (e.g. ABI ViiA7, QuantStudio, Roche LightCycler)
- NanoDrop or equivalent

ASSAY SETUP

- Suggested seeding: 6-well plates at a density reaching **70–90% confluence at harvest**
- **Harvest time:** **6–12 h** for cytokine/inflammatory response; **12–24 h** for oxidative/barrier/apoptosis genes
- **Recommended genes:** IL-1 β , IL-6, IL-8/CXCLi2, TNF- α , IL-10, TGF- β , iNOS, Nrf2, HO-1, NQO1, SOD1, SOD2, CAT, GPx, ZO-1, OCLN, CLDN1/CLDN3, Bax, Bcl-2, caspase-3, GAPDH or β -actin

PROCEDURE

- 1 Seed cells in 6-well plates; allow attachment overnight
- 2 Pretreat cells with samples at 1, 5 and **10 µg/mL** for **12 h**
- 3 Challenge with sterile CP challenge material for the selected period
- 4 Remove medium and wash cells once with PBS
- 5 Add **1 mL TRIzol** per well of a 6-well plate, or the volume recommended by the RNA kit
- 6 Extract total RNA per reagent instructions
- 7 Quantify RNA and assess purity using **A260/A280**
- 8 Reverse transcribe equal RNA input per sample — Luo et al. used **1 µg total RNA**
- 9 Prepare **20 µL qPCR reaction** (Luo et al.): **10 µL** 2× SYBR Premix Ex Taq + **0.4 µL** forward primer + **0.4 µL** reverse primer + **2 µL** cDNA + **7.2 µL** nuclease-free water. (*With a combined primer mix: 0.8 µL primer mix + 6.4 µL water.*)
- 10 Run qPCR using primer-validated cycling conditions. Mei et al. used: **95 °C 10 min → 30 cycles of (95 °C 30 s, 60 °C 1 min, 72 °C 1 min) → final extension 72 °C 8 min**
- 11 Analyze relative expression by **2^{-ΔΔCt}** using GAPDH or β-actin as housekeeping control

READOUT AND ANALYSIS

Protective pattern:

- Lower IL-1β, IL-6, IL-8/CXCLi2, TNF-α and iNOS
- Higher IL-10 / TGF-β
- Higher antioxidant genes
- Preserved ZO-1 / OCLN / claudins
- Lower Bax / caspase-3 and higher Bcl-2

Report fold change vs untreated **and** vs CP challenge-only group.

NOTES / CRITICAL POINTS

- Validate chicken primer specificity and efficiency before the main experiment
- Do not compare gene expression without checking RNA quality and **no-RT controls**

C8 Tight-Junction Immunofluorescence Staining (Optional but Strong)

Source/adaptation — Adapted from Luo et al. (2020), who fixed cells with **4%** paraformaldehyde for **20 min**, permeabilized with **0.5%** Triton X-100 for **20 min**, blocked with **5%** goat serum and stained ZO-1, CLDN12 and OCLN.

Purpose — To visualize whether sample pretreatment preserves epithelial barrier protein localization after CP challenge.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells grown on sterile coverslips
- 24-well plate
- **4%** paraformaldehyde
- PBS
- **0.5%** Triton X-100
- **5%** goat serum blocking solution
- Primary antibodies: ZO-1, occludin and chicken-relevant claudin antibodies (Luo et al. used Bioss anti-ZO-1, anti-CLDN12 and anti-OCLN)
- Fluorescent secondary antibodies (e.g. goat anti-rabbit IgG Alexa Fluor 488 or Cy3)
- DAPI
- Fluorescence microscope

ASSAY SETUP

- Recommended groups: untreated · CP challenge only · sample **10 µg/mL** only · sample 1 + challenge · sample 5 + challenge · sample 10 + challenge
- Use antibody clones validated for chicken if possible

PROCEDURE

- 1 Seed cells on coverslips in 24-well plates; allow monolayer formation
- 2 Pretreat with samples at 1, 5 and **10 µg/mL** for **12 h**

- 3 Challenge with sterile CP challenge material for the selected period
- 4 Wash cells gently with PBS
- 5 Fix with **4% paraformaldehyde for 20 min** at room temperature (Luo et al.)
- 6 Wash **3× with PBS**
- 7 Permeabilize with **0.5% Triton X-100 for 20 min** at room temperature
- 8 Block with **5% goat serum for 20 min** at room temperature
- 9 Incubate with primary antibody diluted in blocking buffer — Luo et al. used **1:300** for ZO-1, CLDN12 and OCLN for **1.5 h** at room temperature
- 10 Wash **3× with PBS**
- 11 Incubate with fluorescent secondary antibody for **1 h protected from light**
- 12 Wash and stain nuclei with **DAPI for 2–5 min**
- 13 Mount coverslips and image with **identical exposure settings**

READOUT AND ANALYSIS

- Protection is supported by preserved junctional ZO-1/occludin/claudin staining compared with disrupted staining in CP challenge-only cells
- Quantify fluorescence intensity or junction continuity using ImageJ

NOTES / CRITICAL POINTS

- Confirm chicken cross-reactivity of antibodies **before purchasing**
- Avoid overfixation if antibody signal is weak

C9 Cytokine ELISA or Multiplex Assay from Chicken-Cell Supernatant

Source/adaptation — Adapted from Jiang et al. (2014), who measured cytokines using a Bio-Plex Multiplex Cytokine Assay, and Choi et al. (2020), who used ELISA from R&D Systems for inflammatory readouts.

Purpose — To quantify secreted inflammatory cytokines after sample pretreatment and CP challenge.

MATERIALS AND REAGENTS

- Cell culture supernatants from chicken epithelial cells or **HD11 cells**
- **Chicken-specific** ELISA kits for IL-1 β , IL-6, IL-8/CXCLi2, TNF- α , IL-10 or TGF- β (R&D Systems, Invitrogen/Thermo Fisher, Abcam, MyBioSource or BioLegend)
- Optional multiplex platform (e.g. Bio-Plex, Bio-Rad)
- 96-well ELISA plate and plate reader
- Protease inhibitor optional for stored supernatants

ASSAY SETUP

- Collect supernatants at **12 h and/or 24 h** after challenge
- Clarify supernatants to remove cells/debris; store at **-80 °C** if not assayed immediately

PROCEDURE

- 1 Treat cells according to the C-group design
- 2 Collect cell culture supernatant into labeled tubes
- 3 Clarify per ELISA kit instructions, or by gentle centrifugation if validated
- 4 Aliquot and avoid repeated freeze–thaw cycles
- 5 Run ELISA or multiplex assay per manufacturer instructions
- 6 Use standards and samples in **duplicate**
- 7 Read absorbance or fluorescence/luminescence per kit/platform
- 8 Calculate cytokine concentrations from standard curves

READOUT AND ANALYSIS

- **Protective pattern:** reduced pro-inflammatory cytokines and/or increased IL-10 / TGF- β in sample + challenge compared with challenge only
- Normalize to cell number or total protein if cell death differs strongly among groups

NOTES / CRITICAL POINTS

- Use **chicken-specific kits** — human/mouse cytokine kits often do not cross-react with chicken proteins

- For HD11 macrophages, Mei et al. supports IL-1 β , IL-6, IL-2, iNOS and IL-10 as relevant endpoints

PART

D

TEM & SEM Imaging Protocols

D1 TEM Image Acquisition and Sample Size Analysis

Source/adaptation — Based mainly on Choi et al. (2020) — **10 μ L** of **50 μ g/mL** sample on 300-mesh copper grids with **2%** uranyl acetate for **5 min** — and supported by Jiang et al. (2014), Nicholas et al. (2017) and Mei et al. (2020).

Purpose — To generate publishable images of CP-samples and quantify vesicle diameter distribution.

MATERIALS AND REAGENTS

- TEM grids prepared in A6
- TEM microscope and EM-core imaging software
- ImageJ/Fiji for size measurement
- Calibration scale from TEM images

ASSAY SETUP

- Image sample-only, PBS blank and mock-sample grids
- Use at least **three independent sample preparations** if possible

PROCEDURE

- 1 Acquire low-magnification images to assess distribution and aggregation
- 2 Acquire higher-magnification images showing vesicle membrane morphology
- 3 Save images with **scale bars**
- 4 Open images in ImageJ/Fiji and set scale using the scale bar
- 5 Measure diameter of **50–100 clearly defined vesicles** per sample lot
- 6 Export measurements to Excel/GraphPad
- 7 Plot size distribution histogram; report mean $\pm$ SD or median with IQR

READOUT AND ANALYSIS

- Representative TEM image panel

- Sample size range and distribution
- Qualitative purity/aggregation assessment

NOTES / CRITICAL POINTS

- Do not measure overlapped aggregates as single vesicles
- Report negative-stain method and grid type in methods

D2 SEM of Sample-Treated CP Cells

Source/adaptation — SEM added as an imaging endpoint to support antibacterial/biofilm results; TEM-based MV morphology and bacterial characterization were used by Jiang et al. (2014), Nicholas et al. (2017), Choi et al. (2020) and Mei et al. (2020).

Purpose — To visualize CP surface morphology after treatment with 1, 5 and **10** µg/mL sample.

MATERIALS AND REAGENTS

- Sample-treated CP samples from B1, or biofilm samples from B8
- SEM-grade fixative approved by EM core (commonly glutaraldehyde-based)
- PBS or cacodylate buffer per EM core SOP
- Ethanol dehydration series
- Critical point dryer or HMDS drying system
- SEM stubs and conductive carbon tape
- Sputter coater with gold/palladium or equivalent
- SEM access

ASSAY SETUP

- **Groups:** CP only · vehicle · mock-sample · sample 1 · sample 5 · sample 10 · positive antibacterial control
- All pathogen samples must be **chemically fixed/inactivated** per EM-core and biosafety approval before removal from containment

PROCEDURE

- 1 Treat CP samples with sample/control using approved CP assay conditions
- 2 At endpoint, collect cells or biofilm-coated coverslips using approved containment procedures
- 3 Add EM-core-approved fixative in the approved ratio/volume; fix for the approved time
- 4 Confirm sample release requirements with the EM core before transport
- 5 Wash fixed samples with buffer per EM SOP
- 6 Dehydrate through graded ethanol series per EM SOP
- 7 Dry samples using critical point drying or HMDS method
- 8 Mount on SEM stubs using conductive carbon tape
- 9 Sputter coat with conductive metal
- 10 Image multiple random fields at low and high magnification

READOUT AND ANALYSIS

Look for surface roughening, shrinkage, collapse, perforation-like damage, debris, aggregation or biofilm-matrix disruption. Interpret SEM **together with CFU and live/dead data**.

NOTES / CRITICAL POINTS

- SEM shows **morphology, not viability**
- Avoid claiming membrane rupture unless supported by multiple imaging fields and viability/permeability data

D3 SEM of Chicken-Cell Morphology After CP Challenge

Source/adaptation — Built from the epithelial injury logic in Luo et al. (2020), who observed morphological damage, viability loss, LDH release, ROS increase and tight-junction disruption after CPB2 exposure.

Purpose — To visualize whether sample pretreatment preserves chicken epithelial-cell morphology after CP challenge.

MATERIALS AND REAGENTS

- Chicken intestinal epithelial cells grown on sterile coverslips
- 24-well plate
- Sample working solutions
- Sterile CP challenge material or approved live challenge model
- SEM fixative and processing reagents per EM core
- SEM stubs, conductive tape and sputter coating supplies
- SEM access

ASSAY SETUP

- **Groups:** untreated · CP challenge only · sample **10 µg/mL** only · sample 1 + challenge · sample 5 + challenge · sample 10 + challenge
- Use parallel wells for viability/LDH/ROS to connect morphology to quantitative injury data

PROCEDURE

- 1 Seed cells on coverslips; allow monolayer formation
- 2 Pretreat cells with samples for **12 h** at 1, 5 and **10 µg/mL**
- 3 Apply CP challenge material for the selected injury time
- 4 Wash gently with PBS if allowed by the model and EM core protocol
- 5 Fix coverslips with EM-core-approved fixative
- 6 Process through dehydration and drying per EM core SOP
- 7 Mount coverslips on SEM stubs and sputter coat
- 8 Image multiple random fields per coverslip
- 9 Score morphology using a predefined scale, or quantify intact monolayer area if possible

READOUT AND ANALYSIS

- CP challenge should show cell rounding, detachment, surface damage or monolayer disruption if the injury model works
- Sample protection is supported by preserved monolayer and cell surface structure

NOTES / CRITICAL POINTS

- SEM is strongest when **paired with MTT/CCK-8, LDH, ROS and tight-junction immunofluorescence**
- Use identical microscope settings across groups when comparing morphology

SECTION



Data Reporting & Statistics

- Report all sample doses as **µg/mL based on BCA protein concentration**, and state that samples were crude unless further purified
- For each endpoint, report **mean ± SD or SEM** from at least **three independent experiments**
- Use **one-way ANOVA** with Dunnett or Tukey post-hoc tests for multiple groups, or **two-way ANOVA** when comparing sample dose and challenge status
- For qPCR, report **$2^{-\Delta\Delta Ct}$ fold change** and include housekeeping gene validation
- For microscopy, include representative images **plus** quantitative analysis where possible — size distribution, fluorescence intensity, junction continuity, biofilm area or morphological scoring
- **Avoid claiming sample-specific causality** unless mock-sample, sterility and toxin-screen controls support it

APPENDIX



Consolidated Materials and Purchasing List

CATEGORY	RECOMMENDED ITEM / BRAND	USED FOR
EV quantification	Pierce/Thermo Scientific BCA Protein Assay Kit	EV protein normalization
Protein profiling	Bio-Rad SDS-PAGE gels or equivalent; Thermo PageBlue / Coomassie G-250	SDS-PAGE
Toxin screen	Anti-alpha toxin/anti-PlcC, anti-NetB, anti-CPB2 antibodies; compatible secondaries; ECL or BCIP/NBT (VWR)	Western blot
DNA/RNA	TRIzol (Invitrogen or TransGen); bacterial RNA kit; DNase I (Thermo); Solarbio Bacterial Genomic DNA kit	qPCR / PCR
cDNA / qPCR	Takara PrimeScript RT reagent Kit with gDNA Eraser; Takara SYBR Premix Ex Taq or ABI SYBR Green Mastermix	RT-qPCR
Cell viability	DoGen Bio Cell Viability Assay Kit, CCK-8/WST-8, MTT (Solarbio) or Promega assays	Viability
Cytotoxicity	Jiancheng LDH kit, Biovision LDH kit or Promega CytoTox 96	LDH release
ROS	Abcam DCFDA/H2DCFDA Cellular ROS Assay Kit or Solarbio DCFH-DA	Oxidative stress
Mitochondria	Beyotime JC-1 kit, Abcam JC-1 kit or Thermo JC-1 reagent	Mitochondrial membrane potential
Apoptosis	Beyotime, BD, Abcam or Thermo Annexin V-FITC/PI kit	Apoptosis flow cytometry
Cytokines	Chicken-specific ELISA kits (R&D Systems, Invitrogen, Abcam, BioLegend, MyBioSource); Bio-Plex if available	Secreted cytokines
Immunofluorescence	ZO-1, occludin and chicken-reactive claudin antibodies; Alexa Fluor/Cy3 secondaries; DAPI	Barrier integrity

CATEGORY	RECOMMENDED ITEM / BRAND	USED FOR
TEM	EMS 300/400 mesh carbon/Formvar copper grids; 2% uranyl acetate	EV morphology
SEM	EM-grade glutaraldehyde fixative, ethanol series, SEM stubs, carbon tape, sputter coating target	CP/cell morphology
Diffusion assays	Laboratory-validated agar, sterile 6 mm well borer, sterile blank disks, forceps, caliper/ruler	Agar well and disk diffusion
MIC/MBC	Sterile microdilution plates, validated CP broth, recovery agar, sterile diluent, low-binding tubes	Minimum inhibitory and bactericidal concentrations

APPENDIX

B

Suggested Experiment Timeline

DAY	ACTIVITIES
Day -7 to -1	Prepare EV lots; perform BCA; run sterility/no-live-CP check; prepare mock-EV; schedule TEM/SEM
Day 0	Seed chicken cells for 96-well, 24-well and 6-well assays. Prepare CP assay materials per approved SOP
Day 1 (morning)	Sample pretreatment of cells at 1, 5 and 10 $\mu\text{g/mL}$. Start CP growth, agar well diffusion, disk diffusion and MIC assays per laboratory SOP
Day 1 (evening) – Day 2	Apply CP challenge material to pretreated cells. Collect bacterial growth/CFU samples. Read diffusion assays; perform MBC recovery from MIC wells as scheduled
Day 2	Run viability, LDH, ROS and/or JC-1 assays. Harvest samples for RNA. Fix samples for IF/SEM as needed
Day 3–5	Run qRT-PCR, ELISA/multiplex, microscopy imaging and data analysis

APPENDIX

C

References

C.1 Primary Literature

- 1 **Jiang, Y., Kong, Q., Roland, K. L., & Curtiss, R. III.** (2014). Membrane vesicles of *Clostridium perfringens* type A strains induce innate and adaptive immunity. *International Journal of Medical Microbiology*, **304**, 431–443.
- 2 **Nicholas, A., Jeon, H., Selasi, G. N., Na, S. H., Kwon, H. I., Kim, Y. J., Choi, C. W., Kim, S. I., & Lee, J. C.** (2017). *Clostridium difficile*-derived membrane vesicles induce the expression of pro-inflammatory cytokine genes and cytotoxicity in colonic epithelial cells in vitro. *Microbial Pathogenesis*, **107**, 6–11.
- 3 **Luo, R., Yang, Q., Huang, X., Yan, Z., Gao, X., Wang, W., Xie, K., Wang, P., & Gun, S.** (2020). *Clostridium perfringens* beta2 toxin induced in vitro oxidative damage and its toxic assessment in porcine small intestinal epithelial cell lines. *Gene*, **759**, 144999.
- 4 **Choi, J. H., Moon, C. M., Shin, T. S., Kim, E. K., McDowell, A., Jo, M. K., Joo, Y. H., Kim, S. E., Jung, H. K., Shim, K. N., Jung, S. A., & Kim, Y. K.** (2020). *Lactobacillus paracasei*-derived extracellular vesicles attenuate the intestinal inflammatory response by augmenting the endoplasmic reticulum stress pathway. *Experimental & Molecular Medicine*, **52**, 423–437.
- 5 **Mei, C., Sun, A. H., Blackall, P. J., Xian, H., Li, S. F., Gong, Y. M., & Wang, H. J.** (2020). Component identification and functional analysis of outer membrane vesicles released by *Avibacterium paragallinarum*. *Frontiers in Microbiology*, **11**, 518060.
- 6 **Balouiri, M., Sadiki, M., & Ibnsouda, S. K.** (2016). Methods for in vitro evaluating antimicrobial activity: A review. *Journal of Pharmaceutical Analysis*, **6**, 71–79.
- 7 **Athman et al.** (2005); **Guo et al.** (2015); **Zaffuto et al.** (2008) — source methods for intestinal cell disaggregation and isolation (Chapter 3).

C.2 Standards and Guidance

- 1 **Clinical and Laboratory Standards Institute (CLSI).** M02: Performance Standards for Antimicrobial Disk Susceptibility Tests. Current edition.
- 2 **Clinical and Laboratory Standards Institute (CLSI).** M07: Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically. Current edition.

- 3 **World Health Organization.** *Laboratory Biosafety Manual*, 4th edition.
- 4 **Centers for Disease Control and Prevention & National Institutes of Health.** *Biosafety in Microbiological and Biomedical Laboratories*, 6th edition.
- 5 **Organisation for Economic Co-operation and Development.** *Principles of Good Laboratory Practice*.
- 6 **ISO/IEC 17025:2017.** General requirements for the competence of testing and calibration laboratories.
- 7 **FAIR Guiding Principles** for scientific data management and stewardship.
- 8 **Institution-specific** biosafety, chemical-safety, equipment and waste-management SOPs.

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